SETTING UP AN ENTERPRISE NETWORK

BY

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Introduction to Computer Networks

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ACKNOWLEDGEMENT

...understanding network performance is more art than science. There is little underlying theory that is actually of any use in practice...

~ Andrew S. Tanenbaum

...and to utterly and completely bastardise Andrew Tanenbaum’s quote, a network is akin to an eldritch godly abomination. Do not try to understand this wretched fiend less you lose all your sanity; if it works, it works – do not question it. Dread it, run from it... you are but at the mercy of its whims.

to sleepless nights and frantic days

ABSTRACT

The following report discusses the setting up of an intelligence-based enterprise network. The network simulation software named 'Cisco Packet Tracer' has been used to design and configure the network of the company based on the requirements and research provided in the report. The report will show how Cisco Packet Tracer can simulate realistic operations of a network; it consists of virtual versions of real networking devices that can be configured using their own CLI interface and it also shows how packets can travel in the network step by step. The departments that have been decided upon will work collectively in ensuring efficient operations, private and secure communications between departments, strong protection and counter measures against cyber threats and a constant improvement in the network to achieve the company's objective. After designing the network, Cisco Packet Tracer allows users to send ping signals (test packets) from one device to another to check if the connectivity of the network performs well. The demo provided along with the report shows this test being performed as proof that the network of the company works efficiently. Therefore, this study will show how helpful network simulators can be in the implementation of an entire network system.

INTRODUCTION

An enterprise network is defined as the IT infrastructure that enables the users, systems, devices and applications within an organisation to connect and exchange data. As the business world becomes more digitised with further technological advancements, it is essential that organisations establish enhanced enterprise networking systems to allow sharing of information and resources and delivery of services worldwide. Nowadays, it is also imperative for businesses to use the latest technologies when setting up or improving their computer networks so that they can benefit from better cybersecurity, increased storage capacity, proper connectivity with real-time voice and video for optimised user experience, easier network management and enhanced readiness for scaling up if needed in the future.

This report focuses on setting up an enterprise network for Eyeton, a new company which will be based in Mauritius. The company will specialise in investigating cybersecurity threats and incidents in other Mauritian businesses as well as support criminal investigations. It is expected to process a large amount of information to assess businesses' vulnerabilities to cyberattacks and use advanced monitoring and intelligence measures to offer cybersecurity solutions to organisations based on their specific needs. It will also support the Mauritian government in the investigation of cybercriminals by performing forensic assessments to understand the type and impact of security breaches. In total, the company will have 300 employees. The company will have several departments namely HR, Health, Finance, Legal, System Administration, IT Support, Research, Operations, Intelligence and Security Analysts. The specific requirements for each department will be discussed further below.

AIMS AND OBJECTIVES

The aim of this report is therefore to establish an enterprise network for this company while taking into account its specific requirements based on the work it will perform. The objectives will include to:

- Understand the network requirements of each department to determine the network need of the entire business
- Establish the specifications and suitability of the network infrastructures
- Design and test a network architecture for the company

ANALYSIS

NETWORKING REQUIREMENTS

The network will be segmented into **12 sub-networks** to host the 9 departments within the enterprise, which will be calculated from the assigned IP address range, using the subnetting process. The IP address will have to also undergo **configuration**.

With the differing amount of traffic that will pass through each network, the different types of **switched Ethernet** will have to be investigated to come to a consensus as to which will be used for each network according to its appetite.

To connect all the logical endpoints and various devices together, the **topology** of the network will have to be decided upon.

Network devices, and their numbers, thereof, will have to be implemented for the good-process of the networking traffic, and are as such:

- **Routers**
- **Switches**

Due to the contemporary nature of the networking and communication domain, wireless communication has impregnated our current zeitgeist, and as such, provisions have to be made to accommodate this type of communication, with devices such as **Wireless Access Points**.

SECURITY DEPARTMENTAL SUBNETWORK AND POLICIES.

"The Art of War teaches us not to rely on the likelihood of the enemy not coming but on our readiness to receive him. Not on the enemy not attacking but on the fact that we have made our positions unassailable"

~ Sun Tzu

With EyeTon IA vying for adherence to **ISO27001**, security is paramount in the setting up of its enterprise network, and as such, various policies, technologies and controls will have to be put in place to ensure strict compliance with these governance structures, *id est*:

- **Alarms with integrated motion detectors** on the outskirts of the building, as well as in every working and living area, of which will amount to:
  - 16 alarm systems
- **CCTV cameras** for the monitoring of movement and operations throughout the building, of which will amount to:
  - 36 CCTV cameras
- A **Defence-in-Depth** (DiD) strategy to ensure redundancy and a multi-layered approach in Threat Mitigation.
- The implementation of **Honeypots** along the network to divert incoming malicious traffic away from the valuable assets.
- The implementation of **Network Segmentation** to segregate devices that need not to communicate with each other, and creating choke points to mitigate the chance of any malicious traffic
- The implementation of **Firewalls** for:
  - filtering ingress and egress traffic,
  - inspecting the said ingress and egress traffic
- The implementation of a **De-Militarized Zone** to quarantine potential attacks within their breached network segments.
- **Permutation and Substitution boxes** will be attached at each of the network devices to ensure traffic in encrypted.

HR, HEALTH, FINANCE AND LEGAL DEPARTMENTAL SUBNETWORK.

The HR department of this organization should not be a complex one. However, it should be secured properly as this consists of data about the staff/patients involved there. Considering this, the number of endpoints is around 8-10 with the weight of the traffic being reasonably light. Due to low number of end users, a small subnet will be used. A database server is required for keeping records of all details involving employees and patients and mail servers for sending and receiving emails from possible recruitments to the organization. The practice of access controls, encryption of data, auditability (logging and monitoring) of user actions, defense in depth and backups should be highly implemented to keep sensitive data secured.

The Finance department is also a small department so around a maximum number of 20-25 endpoints will be needed, resulting in a small subnetwork configuration. The weight of the traffic would be light-moderate depending on peak times of data processing, large transactions, total salary processing, etc. A financial database server is required for storing data involving transactions, salaries, etc. Recommended usage policies are role-based access controls (managers/staffs/other authorized users), separation of duties to avoid single person from having too much control, ensure auditability to check for suspicious financial frauds.

RESEARCH, OPERATIONS AND INTELLIGENCE DEPARTMENTAL SUBNETWORK.

Research Department

The research department is the department where various kinds of testing are carried out. All those analysis data are confidential and require a lot of storage space, so an encrypted cloud storage will be used. The number of endpoints should be around 25 - 30 and the weight of the network traffic should be high considering there is a lot of data transmission happening within the department.

Operation Department

The operation department performs several tasks like cyber breach investigations, ethical hacking, offer cybersecurity solutions to other businesses and assist in the investigation of national cybercrimes. Therefore, the number of End points will be 80 – 90 and the weight of network traffic would be high.

Intelligence Bureau Department

This department is responsible for collecting and classifying all sorts of data and information as well as seeking new strategies to benefit the organization. Hence a maximum number of 30 – 35 endpoints will be needed, and the weight of the traffic would be high due to huge amount data being processed. A database server is needed to keep all the classified information organized.

All the 3 departments above have some requirements in common. Firstly, their end user devices will be PCs, laptops, tablets, printers, and CCTV cameras. Secondly, a DHCP server is required to assign IP addresses automatically. Other servers such as DNS server, web server and ISP server are essential to provide internet access. The departments also practice usage policies such as role-based access control and auditability, thus minimizing the risk of insider attacks.

SYSTEM ADMINISTRATION, IT SUPPORT AND SECURITY ANALYSTS DEPARTMENTAL SUBNETWORK.

The System Administration, IT Support and Security Analysis departments will be on their **own dedicated networks** due to the nature of their sensitive work. This network will be further segmented so as to adhere to the **Defence in Depth policy**, with the proper security protocols and technologies applied throughout because of the low risk tolerance of these positions.

The workforce in these departments will evidently be afforded a greater list of privileges due to the surgical aspect of their work, with:

- IT support needing to be able to assist the end-users on their logical issues, thus needing access privileges to the concerned devices,
- System administrators needing to be able to manage and maintain the organisation’s IT infrastructure, thus needing access privileges to those concerned devices,
- Security analysts needing to be able to prevent and respond to cybersecurity threats and tackle any cybersecurity issue that may arise, thus needing the appropriate access privileges on the network.

Those departments will also benefit from traffic priority from a **QoS** stand-point.

The teams will be comprised of:

- 15 IT support specialists
- 5 system administrators
- 10 security analysts

NETWORK DESIGN AND SPECIFICATIONS

BANDWIDTH

Bandwidth refers to the maximum amount of data that can be transferred from one point to another, that is, uploading and downloading, at a given time. It is normally measured in megabits per second (Mbps) or gigabits per second (Gbps). The bandwidth required for the company depends on a number of factors including number of users and devices connected to the network, the types of applications being used and the number of online activities. In this company, out of the 300 employees, we can assume that the number of active users that will be connected to the network at the same time will be about 80% of the total, that is, 240. This also implies that at least 240 devices including laptops, tablets and work mobile phones will be connected to the network. Different departments will also have different bandwidths needs. For example, the HR department will be mostly carrying out basic tasks such as emails and web browsing, which require low bandwidth of 0.5-1 Mbps per user. Employees in the Health and Legal departments will be frequently using applications such as video-conferencing tools, Office 365 and other cloud-based systems, which normally consume about 1-3 Mbps per user. On the other hand, higher bandwidth demands (equal or greater than 5 Mbps per user) are required by departments such as Operations, Research, Intelligence and Security Analysts among others, where intensive tasks including processing and transmitting of large amounts of data, performing analytics monitoring online, downloading frequent updates and establishing secure channels to connect with client organisations will be performed.

The bandwidth requirement is estimated as follows by considering a list of activities that the company is expected to perform and taking into account the number of users that will be carrying out these tasks simultaneously. As such, it is estimated that the bandwidth requirement of the company is 2.775 Gbps.

Activities	Number of active users at the same time	Bandwidth per user	Total Bandwidths (Mbps)
Basic office applications	0.5	240	120
Video conferencing	2.5	100	250
Emails	1	200	200
Web browsing	1	200	200
Large file transfers	5.5	150	825
Data analysis	6	150	900
System Updates	5	80	400
Total Bandwidth for the company			**2775 Mbps**

SUBNETTING

Subnetting not only allows to elastically expand a network, but also works in favour of security by segregating the network into smaller segments. This is done by splitting the block of assigned IP addresses into several parts, each part representing a distinct internal network, with the sum of them all acting like a single network to the outside world (Tanenbaum, 2003).

ETHERNET

Ethernet is the cable technology used to connect logical devices in a Local Area Network, with **Switched Ethernet** having replaced the original form of Ethernet, aptly named Ethernet, and rendered it obsolete. Switched Ethernet make use of devices called switches to connect the different logical devices' communications cables, forwarding the traffic to and fro (Tanenbaum and Wetherall, 2011).

NETWORK DEVICES

The following are details and specifications of the main **possible devices** to be used in the suggested network design with the aim of satisfying the requirements of the enterprise. These include routers, access points, LAN controllers and switches.

ROUTERS

Cisco CGR1240

Type:	Mid-range ISR Router.
Throughput:	Variable throughput that supports medium to high traffic networks.
Interfaces:	**2 Gigabit Ethernet ports** with speeds of up to 1 Gbps (1000 Mbps).
	4 Fast Ethernet ports with speeds of up to 100 Mbps.
Security:	strong firewalls and VPNs, support for advanced routing protocols and allows WAN/LAN expansion.
Best Use:	is for large departments with high number of endpoints and high traffic regularity. (Main operations)

Cisco 829

Type:	Ruggedized ISR Router design for small, remote and harsh environment with a focus on durability.
Throughput:	50 to 200 Mbps.
Interfaces:	**4 Gigabit Ethernet ports** (throughput of GE ports limited by the performance of the router)
Security:	built in firewall and VPN and support for basic routing protocols.
Best use:	is for small departments with low endpoints and those with light/medium traffic.

Cisco 819HG-4G-IOX

Type:	It is a hardened and compact router, an improvement of the Cisco ISR 819 series. HG for a hardened version (that it is more durable against harsh environment), 4G shows that it has integrated LTE (4G Cellular) support, IOX indicates that it supports the Cisco IOx framework.
Throughput:	50 to 100 Mbps
Interfaces:	It has 4 FE ports and 1 GE port.

Security:	include firewall, Intrusion Prevention Systems (IPS) and a strong application for encryption (e.g. AES, DES, 3DES)
Best use:	is for machine-to-machine applications, industrial automation (e.g. factories, substations), Remote Branch Offices

Cisco 1941 ISR

Type:	An ISR router of the 1900 series designed for small-medium companies.
Throughput:	up to 250 Mbps.
Interfaces:	2 GE ports.
Security:	include strong VPN capabilities, Quality of Service (QoS) for prioritizing different traffic types, good scalability and integrated firewall and IPS.
Best use:	is for small-medium enterprises that need secure connections of all endpoints.

Cisco 4321

Type:	An ISR router that is part of the Cisco ISR 4000 Series.
Throughput:	up to 100 Mbps.
Interfaces:	2 GE ports.
Security:	include Zone-based firewall, VPN, Strong encryption (AES, 3DES), use of ACLs (Access Control Lists) for traffic filtering, IPS and QoS.
Best use:	for small branch offices with moderate traffic (for a higher traffic branch, higher series is most suitable e.g. 4331)

CISCO 4331 will be used

WIRELESS DEVICES

AcessPoint-PT-A

Model:	PT-A of the PT Series Access Points
Type:	Basic standalone access point
Wi-Fi Standard:	802.11b with up to 54 Mbps of Maximum speed

Frequency Band:	5 GHz
Best use:	for SOHO networks (Small Office/Home Office networks) designed for a small number of users

AcessPoint-PT-N

Model:	PT-N of the PT Series Access Points
Type:	basic standalone access point
Wi-Fi Standard:	802.11n (Wi-Fi 4) with up to an approximate speed of 600 Mbps
Frequency Band:	2.4 GHz/5 GHz
Best use:	For medium Home/Office networks that need higher speed that what PT-A can provide

AcessPoint-PT-AC

Model:	PT-AC of the PT Series Access Points
Type:	Standalone indoor Access Point
Wi-Fi Standard:	802.11ac (Wi-Fi 5) with speed of 1.3 Gbps up to several Gpbs
Frequency Band:	5 GHz
Best use:	For a more modern enterprise-level network with a high number of users and a requirement for high-bandwidth.

3702i

Model:	Cisco Aironet 3702i
Type:	Controller-based indoor enterprise-level access point
Wi-Fi Standard:	802.11ac (Wi-Fi 5), with speed of up to 1.3 Gbps
Frequency Band:	5 GHz with dual-band support (2.4 GHz and 5 GHz)
Best use:	For indoor environments of a medium-large enterprise that consists of multiple users.

WLC-3504

Model:	Cisco 3504 of the 3500 wireless controller series.
Type of device:	An enterprise-grade wireless controller.

Purpose:	To provide centralized management for all access points in a network enterprise
Key features:	It can manage a high number of Aps (150-500) while maintaining a centralized configuration.
Best use:	For medium-large enterprises with a considerate number of Access Points

WLC-PT

Model:	Wireless LAN Controller (Packet Tracer version)
Type of device:	A simulation version of a wireless controller in CPT
Purpose:	Used solely for simulation by students/trainees without any real hardware.
Key features:	It has limited Access Point control and has no real throughput
Best use:	In educational related work or testing purposes.

Access Point PT-AC was chosen

SWITCHES

3560-24PS

Model:	Cisco Catalyst 3560-24PS
Type:	Layer 3 switch with Power over Ethernet enabled
Interfaces:	24 FE ports and 2 GE ports
Throughput:	6.5 Mpps (million packets per second)
Security features:	Use of Access Control lists (ACLs), Port Security, user authentications, QoS.
Best use:	In small-to-medium offices that can allow for powering of devices such as IP Phones, Access Points.

3650-24PS

Model:	Cisco Catalyst 3650-24PS
Type:	Layer 3 switch, PoE+ enabled and stackable.

Interfaces:	24 GE ports
Throughput:	40 Mpps
Security features:	ACLs, port security, SNMP, QoS.
Best use:	For enterprises that have as requirement, a high traffic and the need of stacking to ensure redundancy (if one switch fails, others remain operational and maintain the network)

IE-2000

Model:	Cisco Industrial Ethernet 2000
Type:	Layer 2/Layer 3 switch that is ruggedized and is capable of PoE.
Interfaces:	8 FE ports and 2 GE ports
Throughput:	5.66 Mpps
Security features:	ACLs, port security, SNMP, QoS, support for industrial protocols (e.g. STP)
Best use:	Mainly for enterprises built in harsh environments (e.g. Factories, power plants, etc.)

2960-24TT

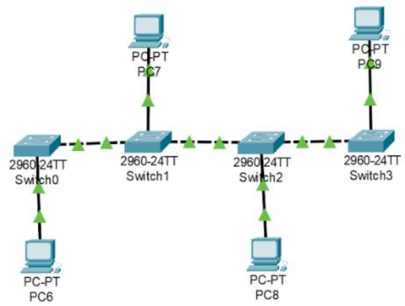
Model:	Cisco Catalyst 2960-24TT
Type:	Layer 2 switch without PoE
Interfaces:	24 FE ports and 2 GE ports
Throughput:	6.5 Mpps
Security features:	ACLs, port security, SNMP, QoS and SNMP management (Simple Network Management Protocol: an application layer protocol that monitors data from network devices to make sure the network operates smoothly)
Best use:	For basic LAN connections of small-medium enterprises.

A combination of layer 2 and layer 3 switches will be used, 3650-24PS and IE-2000

NETWORK TOPOLOGY

BUS TOPOLOGY

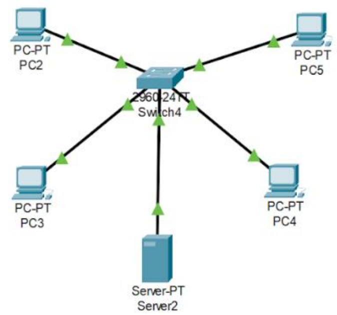
- **Structure**: The end devices are all connected to a backbone cable where transmission of data is taken place. Data travel in both directions and only to the required endpoint.
- **Advantages**: Easy to implement and cost effective, especially for small enterprises.
- **Disadvantages**: A faulty backbone affects all the endpoints, and the rate of data collisions increases as the number of endpoints also increases.



- **Best use**: For a department with low number of endpoints as it satisfies the benefits and minimizes the drawbacks of this topology.

STAR TOPOLOGY

- **Structure**: All end devices are connected to a central hub or switch. Data is transmitted through the hub/switch and then to the destination point.
- **Advantages**: It is easy to add and remove endpoints to the network and it is also more reliable as a faulty cable/endpoint does not affect the whole network, thus improving performance.

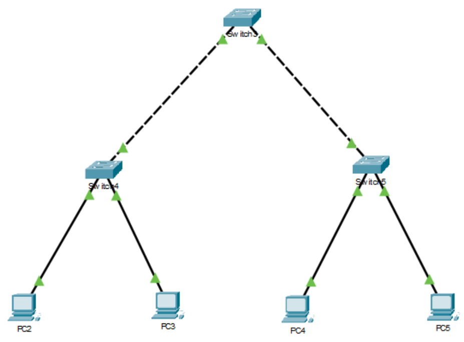


- **Disadvantages**: It can be more costly due to more cables and failure of entire network depends on the main hub/switch.
- **Best use**: For departments with a medium-high number of endpoints

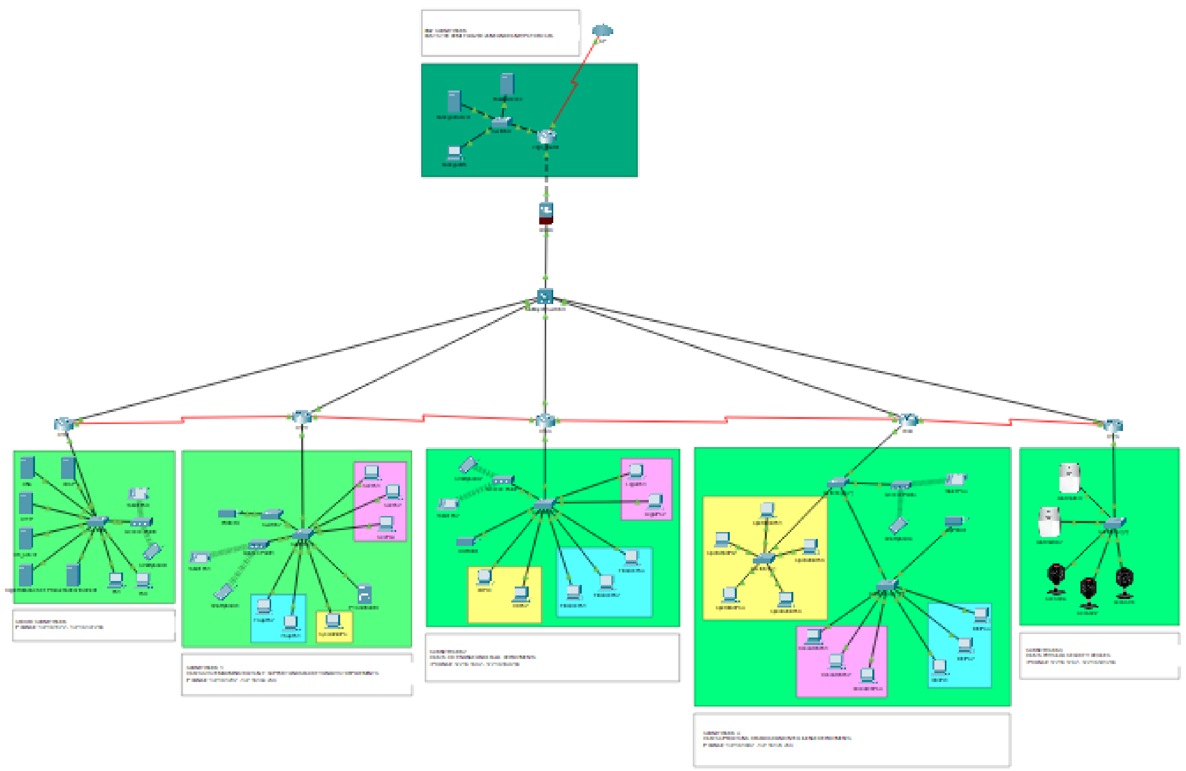
HIERARCHICAL TOPOLOGY

- **Structure:** It is used to connect multiple star-topology network segments together. The network is divided into the Core, Distribution and Access layer, each with its own distinct property. The Access layer (bottom) is where all the endpoints are connected to the network, the Distribution layer is where data from the endpoints is routed throughout the network, and the Core layer is the glue that holds and connects the Distribution layer devices together.

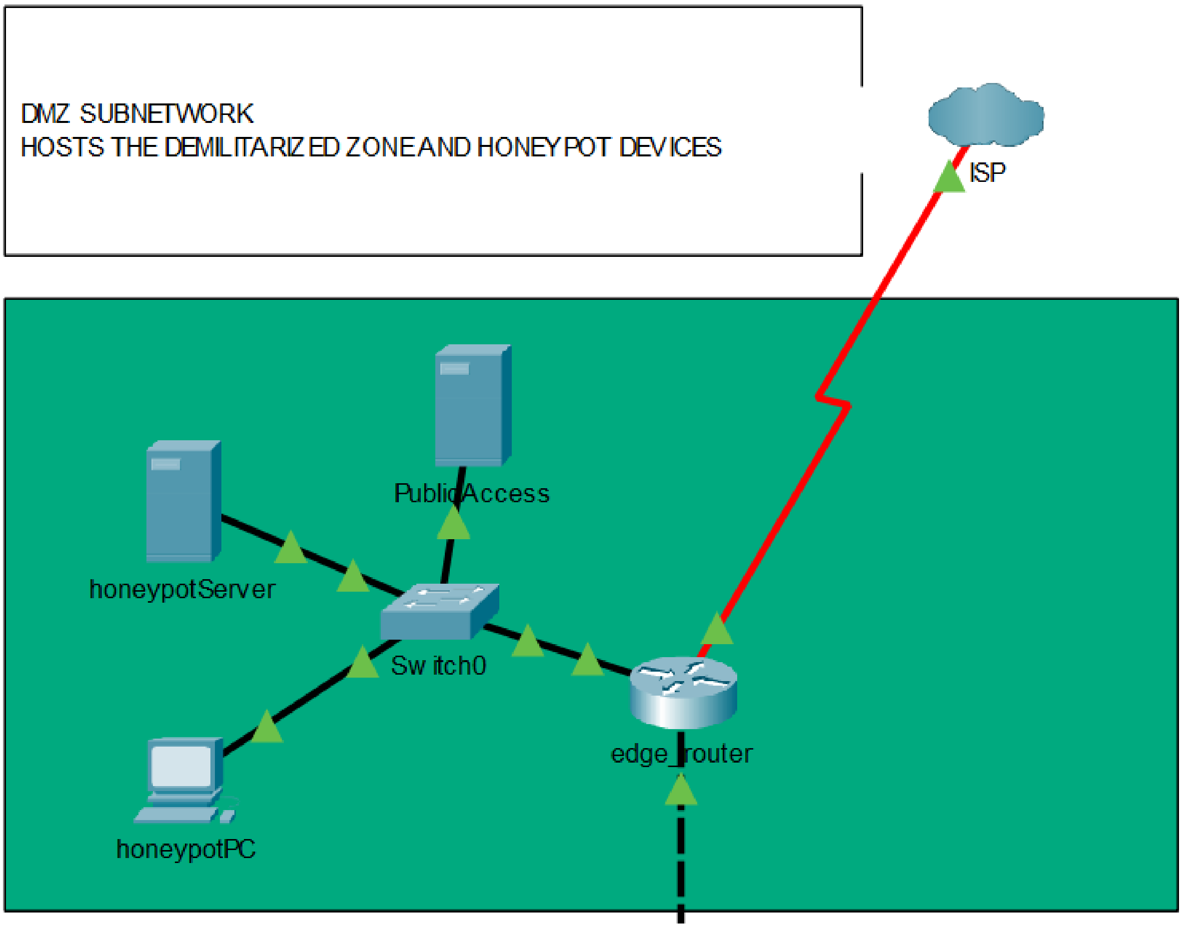
- **Advantages:** Enables for better scalability, divides the network into more manageable chunks for maintenance and troubleshooting as problems are isolated to their own specific layers.
- **Disadvantages:** due to its the number of network devices needed, hierarchical topology makes for a more complex and costly network.
- **Best use:** for large, scalable networks.



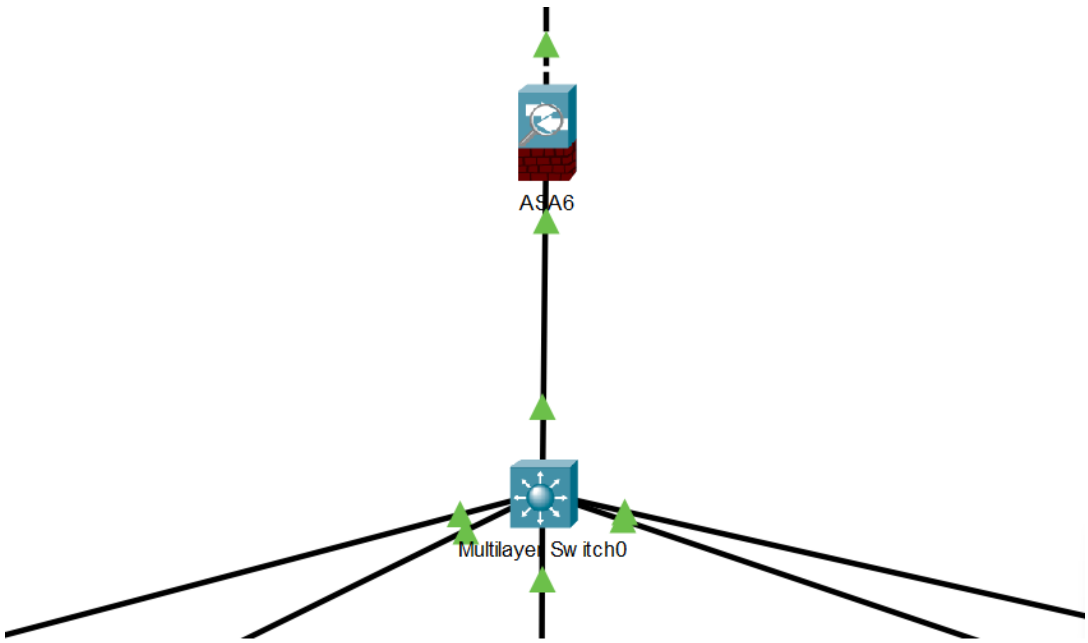
Hierarchical-topology will be implemented, with star-topology at the Access layer



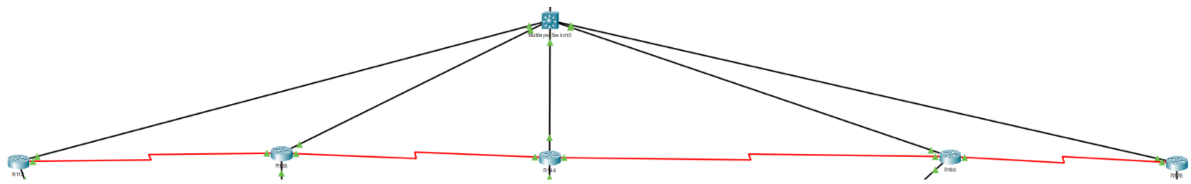
A VIEW OF THE WHOLE NETWORK



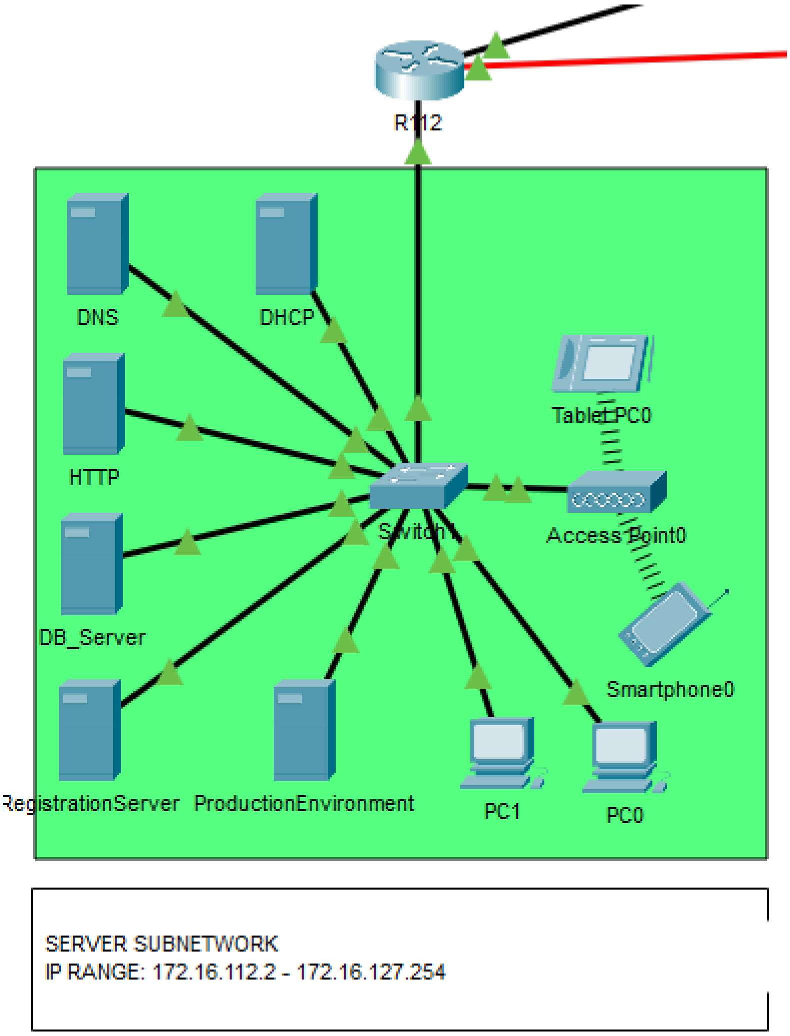
TOPOLOGY OF THE DMZ



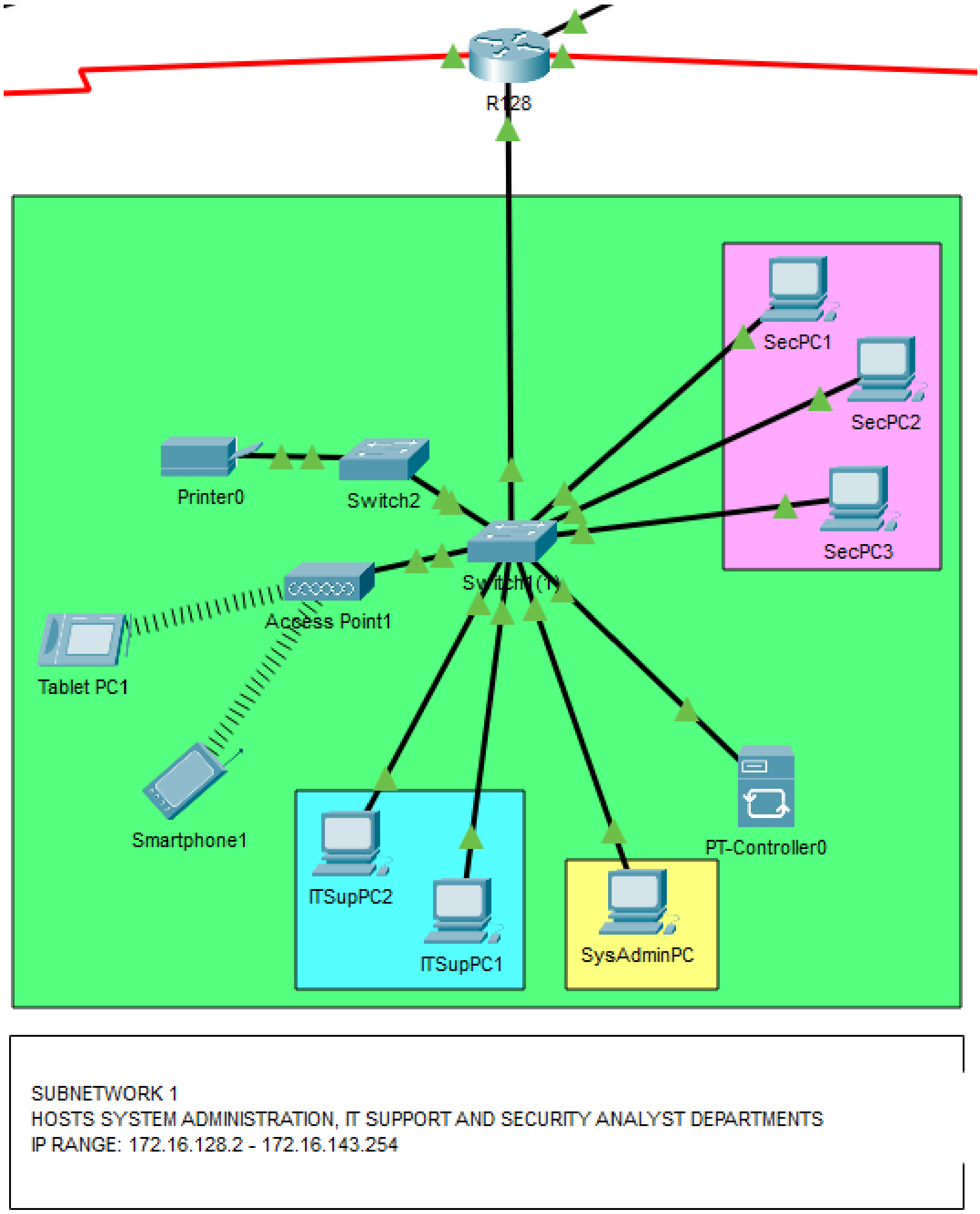
THE FIREWALL (WITH ROUTING CAPABILITIES) AND LAYER 3 SWITCH AT THE CORE LAYER



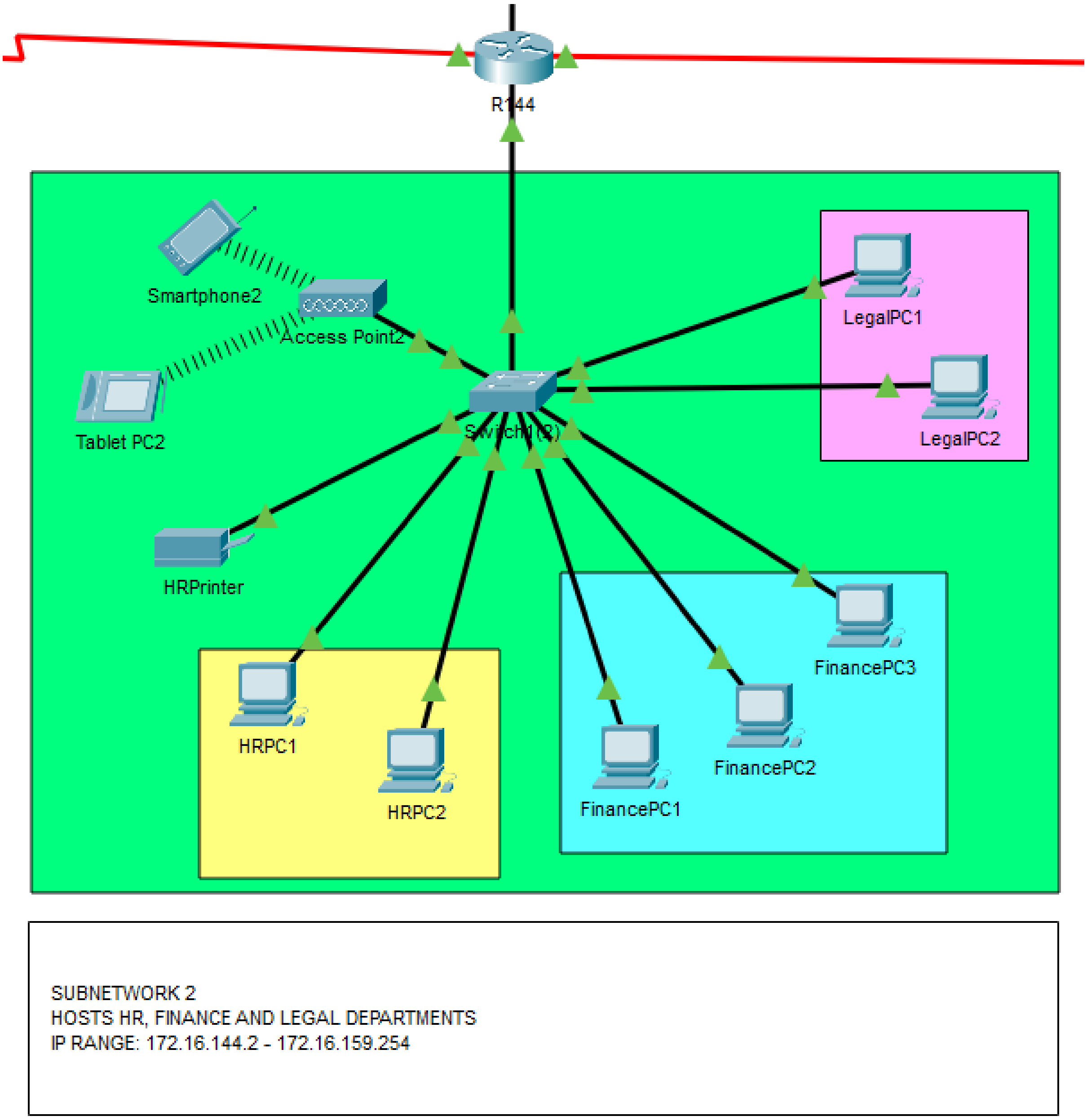
DISTRIBUTION LAYER WITH ROUTERS FOR THE SUBNETS



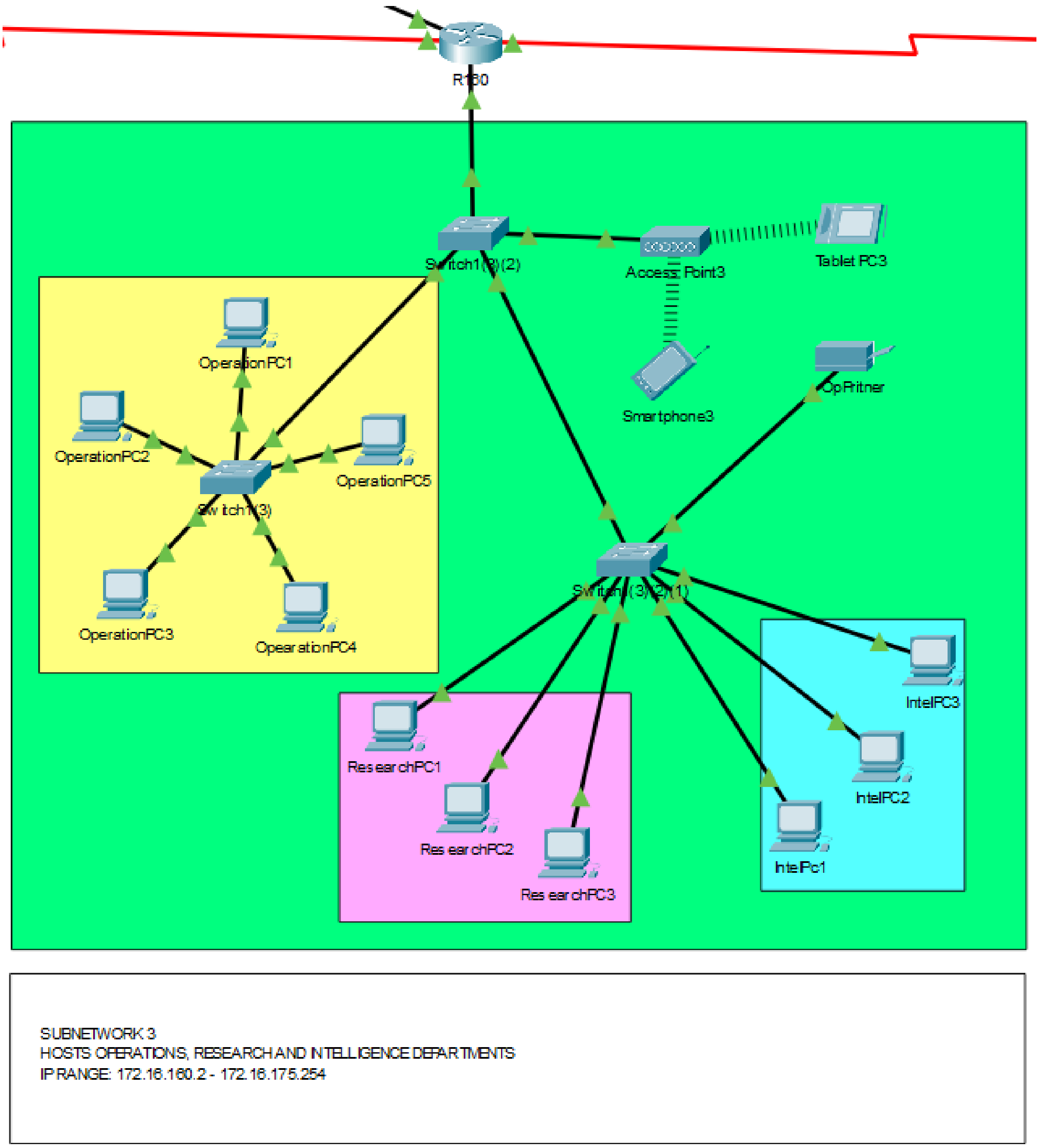
THE SERVER SUBNETWORK AT THE ACCESS LAYER



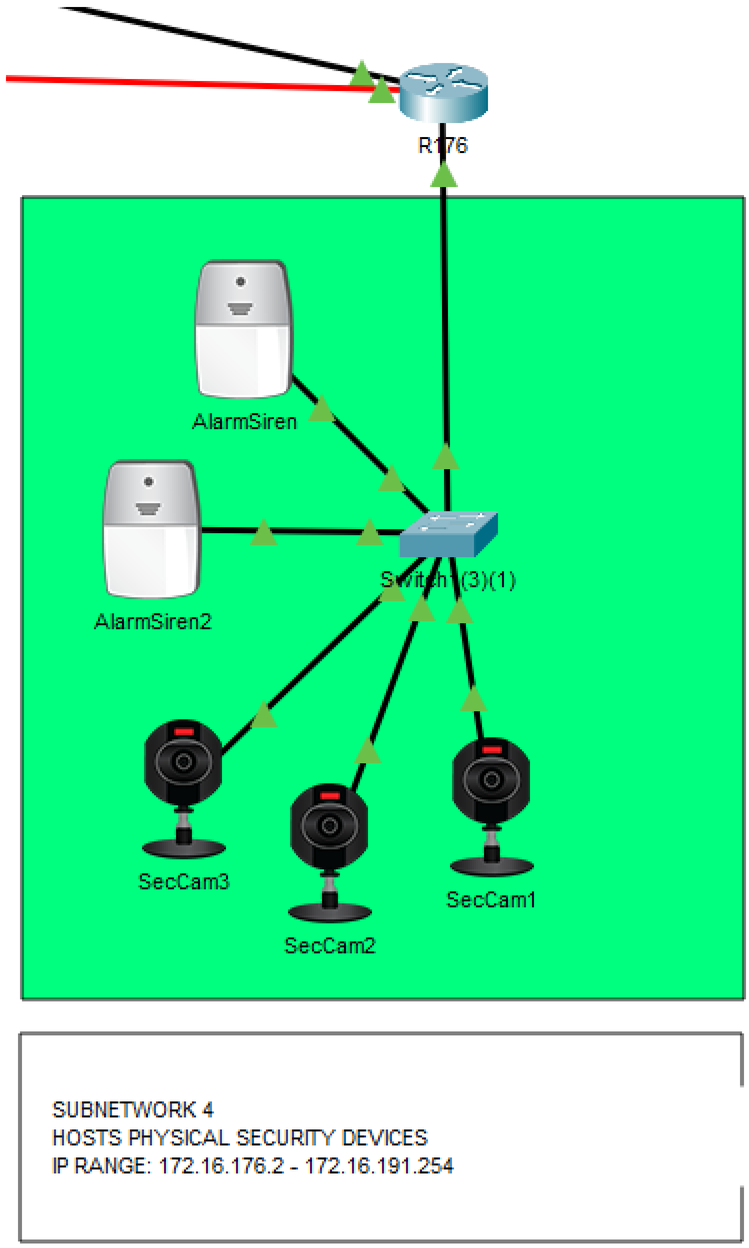
THE SECOND SUBNETWORK AT THE ACCESS LAYER



THE THIRD SUBNETWORK AT THE ACCESS LAYER



THE FOURTH SUBNETWORK AT THE ACCESS LAYER



THE SECURITY SUBNETWORK AT THE ACCESS LAYER

NETWORK SERVERS

DHCP (DYNAMIC HOST CONTROL PROTOCOL) SERVER

A DHCP server is a device implemented within a network to automatically assign IP addresses, subnet mask, default gateway and DNS server address to multiple devices from within its scope. It gives the host devices the right configuration information to carry out **Transmission Control Protocol (TCP)** with other devices.

HOW DOES THE DHCP SERVER WORK?

Modern networking devices have a DHCP client system integrated within its core functionalities by the manufacturers, and this allows them to connect to the DHCP server and request an IP address. Along with the request, the MAC address of that specific device is also sent.

- DHCP servers consist of a (User Datagram Protocol) UDP port number **67**
- DHCP clients consist of UDP port number **68**

When the request arrives at the DHCP server (UDP port number 67), the latter sends an IP address message together with the MAC address it received, to all UDP port number 68 (DHCP clients) on the network but only the device with the matching MAC address is given the authority to use the IP address.

STEPS TAKEN TO OBTAIN AN IP ADDRESS FROM DHCP SERVER:

1. A packet known as **DHCPDISCOVER** is broadcasted on the network by a computer.
2. Upon receiving the packet, the DHCP server answers with a **DHCPOFFER** packet.
3. The device then broadcasts a **DHCPREQUEST** packet.
4. To which, the DHCP server replies with a **DHCPACK** packet. Thus, granting the client device with a **leased** IP address.

ADVANTAGES:

- Less time-consuming since there is no need to configure multiple IP addresses manually.
- Less likely for errors (human errors) to occur since there IP addressing is done automatically.

- IP addresses are leased for a limited amount of time. So, if a host device has been inactive for too long or removed from the network, the IP address returns to the **IP pool** of the DHCP server.

DISADVANTAGES:

- The server may crash or become unresponsive, and no more IP addresses will be allocated, thus new devices won't be able to connect to the network.

DHCP RELAY AGENT

A DHCP relay agent is a network device(router) that allows user devices within one network to make requests for IP addresses from a DHCP server in another network. Broadcast communication between different networks is not feasible and so the DHCP client's DHCPDISCOVER packet will never reach the DHCP server without the use of a DHCP relay agent. The DHCP relay agent as its name suggests relays the request that the DHCP client makes to the DHCP server by attaching the IP address of that server to the message before passing it to the right network interface through unicast communication.

REQUIREMENTS OF DHCP SERVER WITH MODERATE TRAFFIC:

Hardware: -

CPU:	8 core processor
RAM:	12 GB
Storage:	100 GB SSD
Operating System:	64 bit
Network interfaces:	1 gigabit-ethernet network interface or more

Software: -

Ports:	UDP ports 67 and 68 should be available
Microsoft Windows Server OS:	version 2022

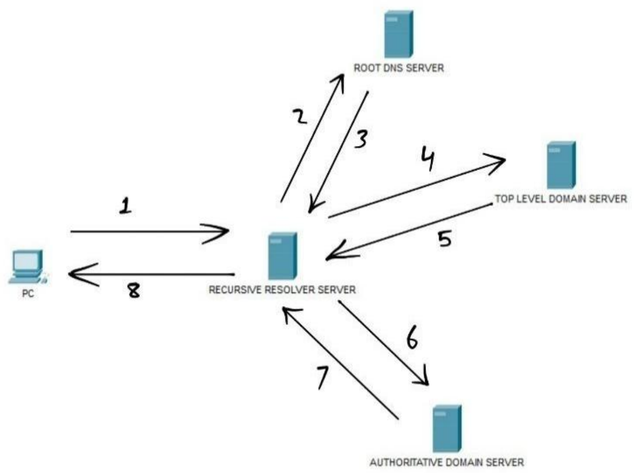
DOMAIN NAME SERVER

DNS server is a networking device that acts as a database server which records the domain names for specific IP addresses. Since humans find it difficult to remember long lines of numbers, IP addresses are classified as domain names within the database such as Google.com, Wikipedia.org etc. Thus, facilitating communication within networks for the client computer to reach websites or even other computers.

EXAMPLES OF DNS SERVERS

- Cloudflare 1.1.1.1
- Google Public DNS
- Quad9

HOW DOES DNS SERVER WORK?



In this scenario, a user attempts to open the Hackathon.com webpage on the web browser. However, its IP address is not found on the cache memory of the PC. The steps below show how DNS queries are carried out to solve this situation: -

1. PC sends a query for the IP address of Hackathon.com to the recursive resolver server.
2. The resolver server searches within its own cache memory to check whether it has the corresponding IP address. Upon getting no result, it will send the query to the root DNS server.
3. The root server then tells resolver server to go to the Top Level Domain (TLD) server that occupies **.com** domains.
4. Resolver server then ask for the IP address on the TLD server.

5. The TLD server tells resolver server to send the query to specific Authoritative domain server.
6. Resolver server sends the query to that authoritative name server.
7. The authoritative name server replies with the IP address of Hackathon.com.
8. Finally, the resolver server gives the PC the IP address to open Hackathon.com.
9. The resolver server saves that IP within its own cache memory, in case it might need it in the future so that it doesn't have to go through all those steps again.

Recursive resolve server – It is in charge of handling the queries of IP addresses from the client device and if it cannot find what it needs from its own cache memory, it will have to perform queries to other DNS servers to reach the desired outcome. The resolver server is usually the local ISP server such as MyT and Emtel.

Root servers – they are the essential servers at the top of the DNS hierarchy. There are 13 sets of such servers around the world, controlled by 12 different organizations. Every one of those servers has a unique IP address. The root servers contain the IP addresses of TLD servers that handle the domains of .com, org, .net, etc.

Top Level Domain (TLD) servers – the TLD servers are one level lower in the hierarchy compared to the root servers and they contain the IP addresses of Authoritative nameservers that handle the specific domains of .com, org, .net, etc.

Authoritative Domain servers – these servers are the last in hierarchy level and they keep track of every IP address of specific domains within their respective zone type. They are divided into 2 categories of servers:

- **Primary Domain Name Server** is organized to store a domain's original DNS zone data.
- **Secondary Domain Name Server** acts like a backup server for the primary server to protect the availability of the data.

ADVANTAGES OF DNS SERVERS

1. It is easier for users to remember domain names instead of the IP addresses.
2. Webpages are often modified and require new IP addresses each time. The DNS server handles that task for us instead us having to do it manually.

DISADVANTAGES OF DNS SERVERS

1. Possibility of a DNS attack to change the IP address of a domain name from real to fake to deceive users.
2. Malicious people can create domain names similar to real ones to fool them into revealing their credentials.

REQUIREMENTS OF DNS SERVER WITH MODERATE TRAFFIC

Hardware: -

CPU:	dual-cores (1.6GHz to 2.5GHz)
RAM:	1 GB
Storage:	2 GB SSD
Network interfaces:	1 gigabit-ethernet network interface or more

Software:

AdGuard Home:	To filter ads and trackers
Un-bound:	Works as a recursive DNS resolver that protects the privacy of a network

WEB SERVER

A Hyper Text Transfer Protocol (HTTP) server is a device that stores website files and when a client device makes a request for a particular webpage, the server will provide it with the files it needs if the request is valid. The web server holds a variety of accessible ports; however, the most commonly used ports are: -

Port 25 – HTTP (for web browsing)

Port 443 – HTTPS (for secure web browsing)

TYPES OF WEB SERVERS

Static web server: this server provides client devices with information that can only be read. (e.g. Apache HTTP Server)

Dynamic web server: the server allows the information contents within database to be updated or removed. (e.g. TEMU, YouTube, etc...)

EXAMPLES OF WEB SERVERS:

1. Apache HTTP server
2. Microsoft Internet Information Services (IIS)
3. NGINX

REQUIREMENTS OF WEB SERVER WITH MODERATE TRAFFIC:

Hardware: -

CPU:	12 core processor
RAM:	16 GB
Storage:	500 GB SSD
Operating System:	64 bit
Network interfaces:	1 gigabit-ethernet network interface or more

Software:

Apache HTTP Server:	to process client devices' HTTP request and respond accordingly.
Microsoft Windows Server OS:	version 2022(to manage software programs)

SECURITY SPECIFICATIONS

GOVERNANCE, STANDARDS AND COMPLIANCE

Governance controls how an organisation makes decisions and influences the behaviours of its employees – it is most closely associated with administrative controls.

Controls form part of all major cybersecurity frameworks, and Eyeton aims to get certified by the International Standards Organisation through its ISO27001 standard; which has various requirements for risk assessment and risk management, to which we will adhere through the network design.

DEFENCE IN DEPTH

DiD is a step away from the traditional cybersecurity defensive posture, away from the monolithic security stance and towards a *motte and bailey* type defence with multiple layers of access controls layered from different levels making sure that a single breach will not cripple the network in its entirety.

The multiple security zones ensures that if one security control is compromised, there will always be another one acting as a *libero* and blocking the way further into the network.

In this way, DiD deters attackers and encourages them to focus on other easier targets.

HONEYPOTS

Honeypots are run-of-the-mill devices that aren't necessary useful to those found within the network. They cannot be communicated with, they cannot be connected to, they cannot be used in any way; but what they provide is a bait and alert as to any security breach.

Honeypots are usually put in areas where users shouldn't be connecting, *id est*, in between the organisation's internal and the outside/public network, and more commonly in the DMZ. If scanned by an attacker, the honeypot behaves like any other device that is to be found on the network, however, they will alert the security team if any connection is made to them, basically acting as a tripwire and alerting of an incoming breach.

NETWORK SEGMENTATION

Complete segmentation occurs when a network is utterly and entirely isolated from all outside communications – but, this is beyond our scope as some form of communication must be enabled between all our subnetworks.

Network Segmentation breaks an organisation's network into smaller and more manageable security zones, controlling the traffic among and between the networked devices. It is an effective way to achieve DiD, and is put in place in networks where people in different roles and security groups need not know about other areas of information/traffic.

FIREWALLS

A firewall is a program or device that uses established security policies to manage incoming and outgoing network traffic. It can be defined as a tool that protects an internal network from an external network that is considered to be not secured, such as the Internet.

To safeguard this company against cybersecurity threats, the Cisco Secure Firewall 4200 Series offer a viable high-end security solution. They consist of features such as quicker traffic decryption, Virtual Private Network (VPN) to enable secure communication, URL filtering which thereby allows administrations to decide which websites to block or allow access to depending on the company policy, Threat inspection for applications through Application Visibility Control (AVC) and Intrusion Prevention System (IPS) that identifies and blocks potential network threats. The Cisco firewalls also include Transport Layer Security (TLS) Hardware Decryption, which implies that the firewall depends on built-in hardware devices to complete the process of decrypting, inspect and re-encrypting encrypted network traffic, hence enabling effective performance. Furthermore, to ensure secure voice communication through trusted devices only, the Cisco firewalls enable traffic to flow through the port 5060, which is normally used for SIP (Session Initiation Protocol) that allow voice calls over the Internet.

The following are the specifications for the different options under the Cisco Secure Firewall 4200 Series:

<u>Cisco Secure Firewall 4215</u>

Use:	For mid-size and large enterprises or headquarters with potential to scale up in future
Throughput:	based on a 1024-byte packet size (including AVC and IPS) <u>65 Gbps</u>
Max. concurrent sessions:	15 million
TLS Hardware Decryption:	20 Gbps Throughput
IPsec VPN Throughput:	(maximum rate of encrypted information that a VPN can transfer and receive over an encrypted channel): <u>45 Gbps</u>
Max. concurrent VPN connections:	20,000
URL filtering:	(number of categories into which the firewall can sort websites) <u>more than 120 categories</u>

<u>Cisco Secure Firewall 4225</u>

Use:	Better suited for data centres
Throughput:	based on a 1024-byte packet size (including AVC and IPS) <u>80 Gbps</u>
Max. concurrent sessions:	30 million
TLS Hardware Decryption:	30 Gbps Throughput
IPsec VPN Throughput:	80 Gbps
Max. concurrent VPN connections:	25,000
URL filtering:	more than 120 categories

<u>Cisco Secure Firewall 4245</u>

Use:	For service providers dealing with high traffic volume
Throughput:	based on a 1024-byte packet size (including AVC and IPS) <u>140 Gbps</u>
Max,. concurrent sessions:	60 million
TLS Hardware Decryption:	45 Gbps Throughput
IPsec VPN Throughput:	140 Gbps
Max. concurrent VPN connections:	30,000
URL filtering:	more than 120 categories

For this cybersecurity company, the option Cisco Secure Firewall 4225 is the most suitable as the company offers services to other client businesses such as monitoring online activity for detecting threats and providing tailored cybersecurity solutions. This implies dealing with large traffic volume and managing quite a few client networks as well as managing and safeguarding its own network. Although Cisco Secure Firewall 4215 appears to be sufficient with a 65Gbps throughput, the Cisco Secure Firewall 4225 provides an additional 15Gbps that is useful to process large traffic volumes and perform sophisticated analytics. Additionally, when setting up a firewall system, it is also important to configure access to port 5060

DEMILITARIZED ZONE

The DMZ is an area isolated from the internal network and is designed to be accessible by outsiders to the organisation – because we do not want to allow outsiders to connect to our internal network for fear of being compromised. It is often host to the public web server, email server, file, and other resource servers.

PERMUTATION AND SUBSTITUTION BOXES

Encryption can be implemented on the physical spectrum itself through simple electrical circuits known as Permutation boxes and Substitution boxes (P-box and S-box). Such devices can be built in the hardware to achieve great encryption speed since encoders and decoders have negligible gate delays. Below, a representation of the devices are shown – and quoting Andrew Tanenbaum, "*…the real power of these basic elements only becomes apparent when we cascade a whole series of boxes to form a product cipher…*" (Tanenbaum and Wetherall, 2011).

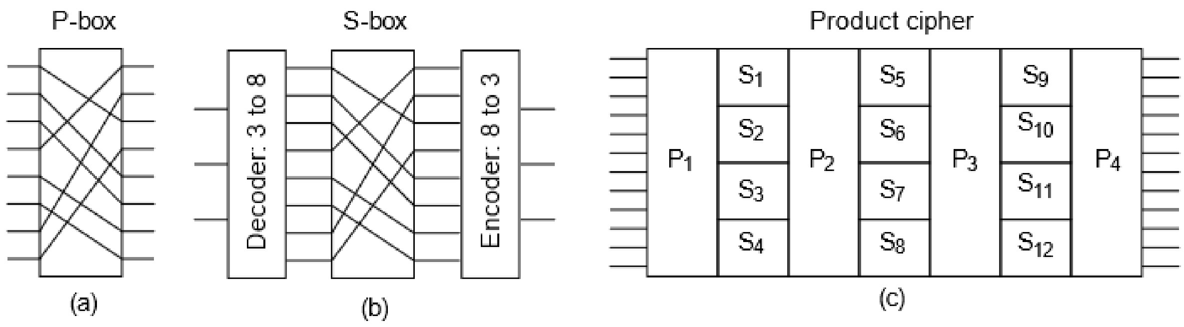


Figure 8-6. Basic elements of product ciphers. (a) P-box. (b) S-box. (c) Product.

USAGE POLICIES

The following usage policies are chosen and refined to meet the standards of the company. Being an intelligence company, it is concluded that the aim should be to have a strong focus on security-based usage policies. This will significantly help in ensuring maximum protection against any security threats, internal or external.

ACCESS CONTROL

They determine who should be granted access to data and information on the network under certain circumstances. They help in preventing unauthorized access and in making sure that only those who are allowed in the system get access to it. Such policies can be implemented through the following techniques: Role-based access, VLAN segmentation with the goal of isolating departments and the use of Multi-Factor Authentication (Exabeam, 2025).

AUDITABILITY

They indicate that all important actions, digitally or physically, should be logged and monitored to be able to detect, analyze and respond to threats. They help in detecting suspicious activities and support any need of investigation when an incident happens. These policies are to be implemented in the form of a centralized log management system to provide regular updates/reviews of all essential activities on the network.

PASSWORD AND AUTHENTICATION

They show the importance of how strong a password should be and how they should regularly be updated. It is a crucial process that all users in the company should put into practice to prevent any data breaches due to unauthorized access. These policies should ensure that users use unique and complex passwords to reduce the risk of commonly used attacks such as brute force. It should also be encouraged to change passwords regularly and to implement biometric passwords for more sensitive data access. Finally, authentication such MFA adds another layer of security to the protection of data (Exabeam, 2025).

PATCH MANAGEMENT

The management of patches involves the correct scheduling and processing of patches and their regular updates. It is a crucial step in fixing the various undetected vulnerabilities that are exploited by cyber criminals. These policies can be effectively implemented through the regular monitoring and testing of patches and thereby conducting any necessary latest updates and changes (Exabeam, 2025).

INCIDENT RESPONSE AND REPORT

This policy is about reacting strategically to cyber threats by being able to detect and respond to them in time. By using the following techniques, these procedures can be helpful in doing so: the use of proper technologies (e.g. monitoring and logging systems) to identify the threat, applying the necessary controls to mitigate any risks and continuously making reviews to prevent the same threat from happening again (Wiz Experts Team, 2025).

NETWORK IMPLEMENTATION

IP CONFIGURATION

Base Network Address: 172.16.0.0

Subnet Mask: 255.255.240.0/20

The subnet mask specifies the range of bits allocated to the network identification and host identification, informing the devices on to which network they are currently residing.

NETWORK ADDRESS	ADDRESS RANGE	USAGE
172.16.0.0	172.16.0.1 – 172.16.15.254	Used for the edge router and devices found in the DMZ.
172.16.16.0	172.16.16.1 – 172.16.31.254	Used for the firewall and internal routers.
172.16.32.0	172.16.32.1 – 172.16.63.254	Used as the port addresses to connect the routers of the different subnets together, and to segregate and segment the network for security purposes.
172.16.64.0	172.16.64.1 – 172.16.79.254	Used as the port addresses to connect the routers of the different subnets together, and to segregate and segment the network for security purposes.
172.16.80.0	172.16.80.1 – 172.16.95.254	Used as the port addresses to connect the routers of the different subnets together, and to segregate and segment the network for security purposes.
172.16.96.0	172.16.96.1 – 172.16.111.254	Used as the port addresses to connect the routers of the different subnets together, and to segregate and segment the network for security purposes.
172.16.112.0	172.16.112.1 – 172.16.127.254	Used for the network hosting the servers.
172.16.128.0	172.16.128.1 – 172.16.143.254	Used for the network hosting the System Administration, IT Support and Security Analyst departments.

172.16.144.0	172.16.144.1 – 172.16.159.254	Used for the network hosting the HR, Health, Finance and Legal deparments.
172.16.160.0	172.16.160.1 – 172.16.175.254	Used for the network hosting the Research, Operations and Intelligence deparments.
172.16.176.0	172.16.176.1 – 172.16.191.254	Used for the network hosting the physical security and access control devices.

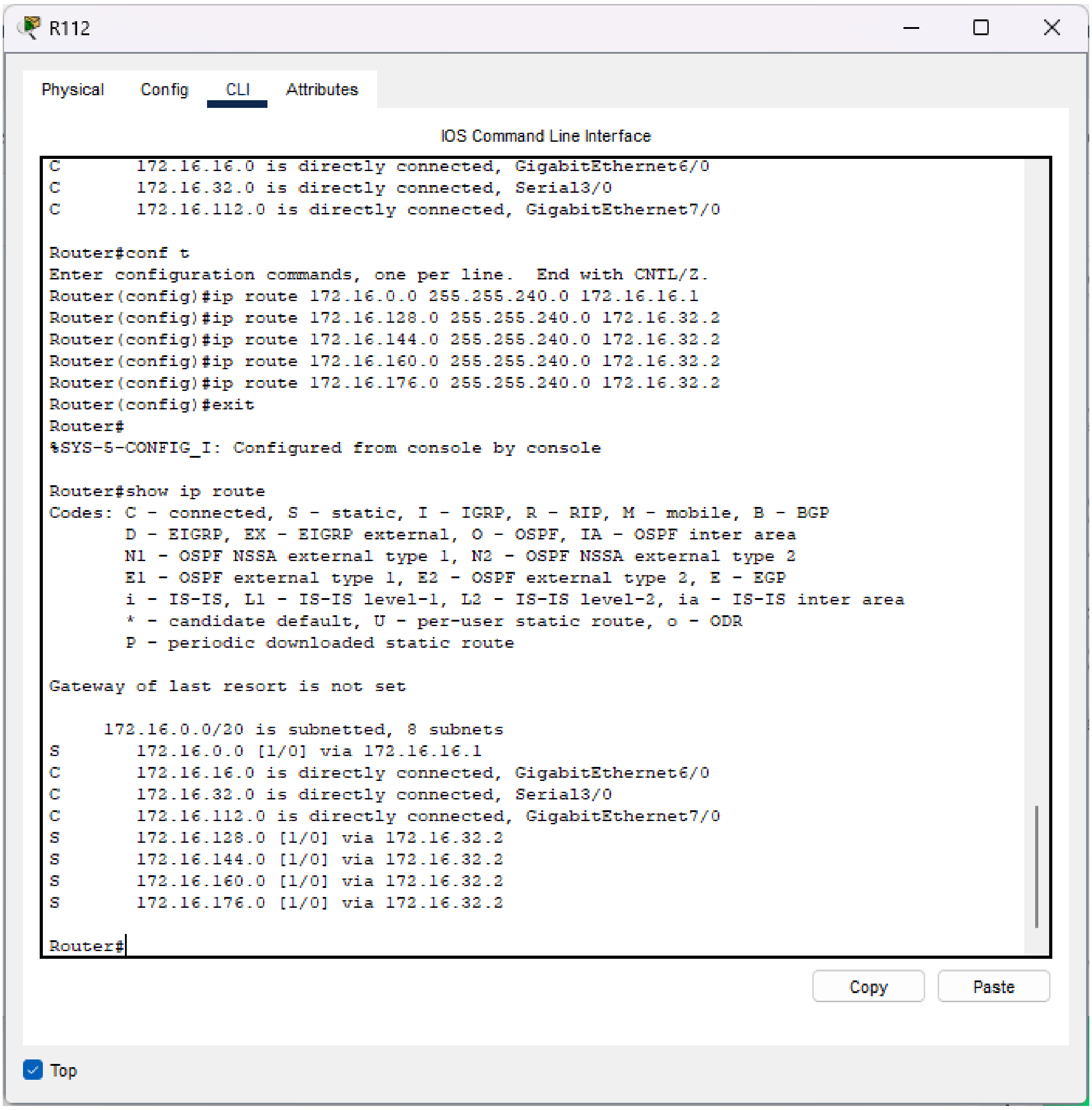
ROUTER CONFIGURATION

IP ROUTING

IP routing is required to allow the different subnets to communicate with each other, otherwise, they would act as if segregated from each other, and the devices connected to each of them would never be able to talk to one another (GeeksForGeeks, 2025).

Exempli gratia: Without IP routing, an HR employee would not be able to access the employee database since they are found on different subnetworks.

What IP routing does is forwarding the packets from a network towards the next hop to reach the desired network.



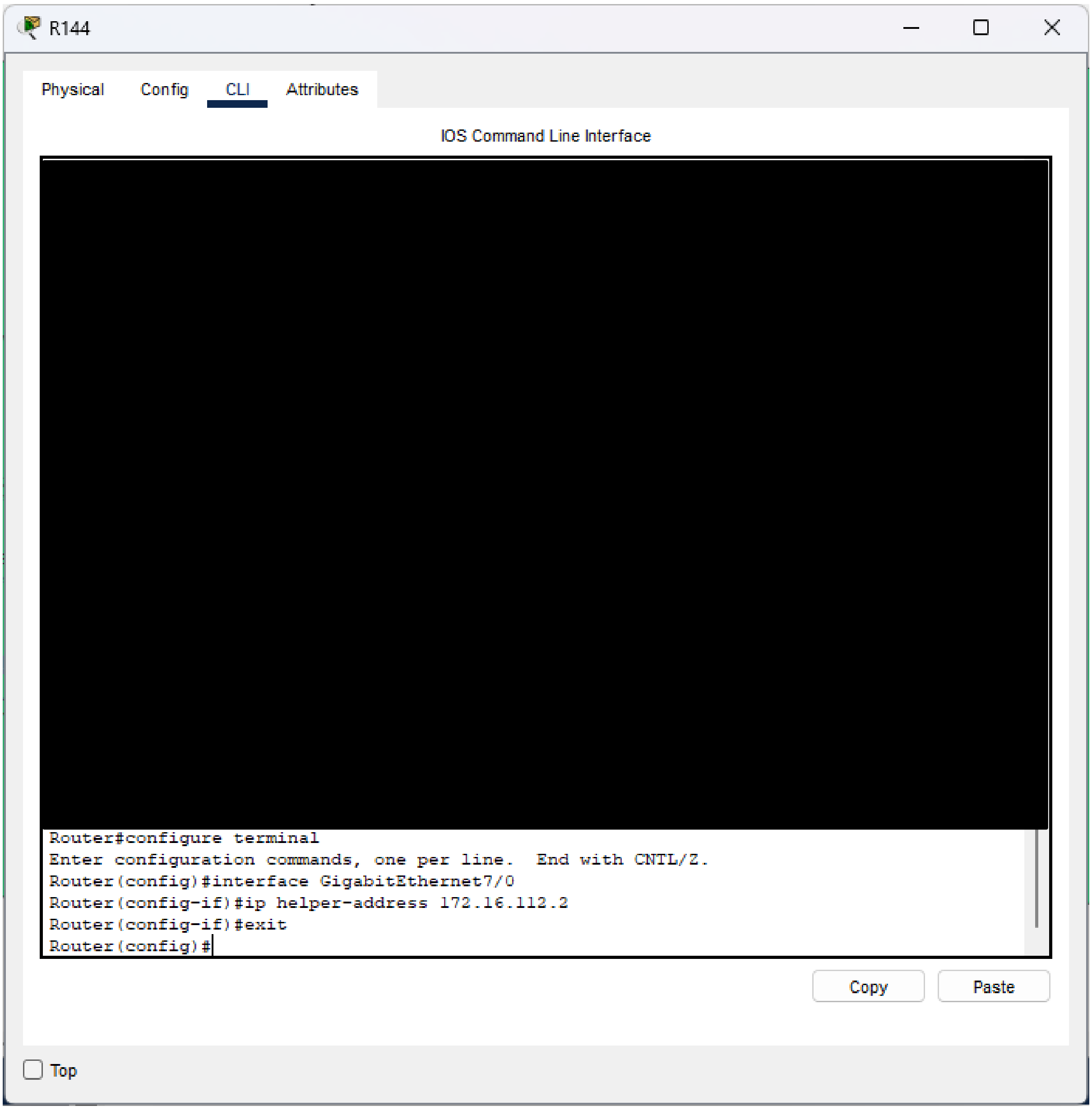
The above displays the commands to apply IP routing to router R112. A breakdown of the commands is:

- `enable`
    - grants root access to the router (not shown here).
- `conf t`
    - makes configuring the router possible.
- `ip route [network address] [network mask] [hop address]`
    - `ip route` enables the configuration of packet forwarding between the networks.

- `[network address]` is the address of the network that is the destination address of the packet.
- `[network mask]` is the address mask of the network address.
- `[hop address]` is the IP address of the next hop (router) in the direction of the network of the destination address of the packet.

DHCP RELAY

In the case of a DHCP server not being found on the same network as the other host devices, DHCP relay needs to be implemented to inform the router of the address of the DHCP server; and hence, the host devices are able to successfully request an IP address from the server (ComputerNetworkingNotes, 2025).

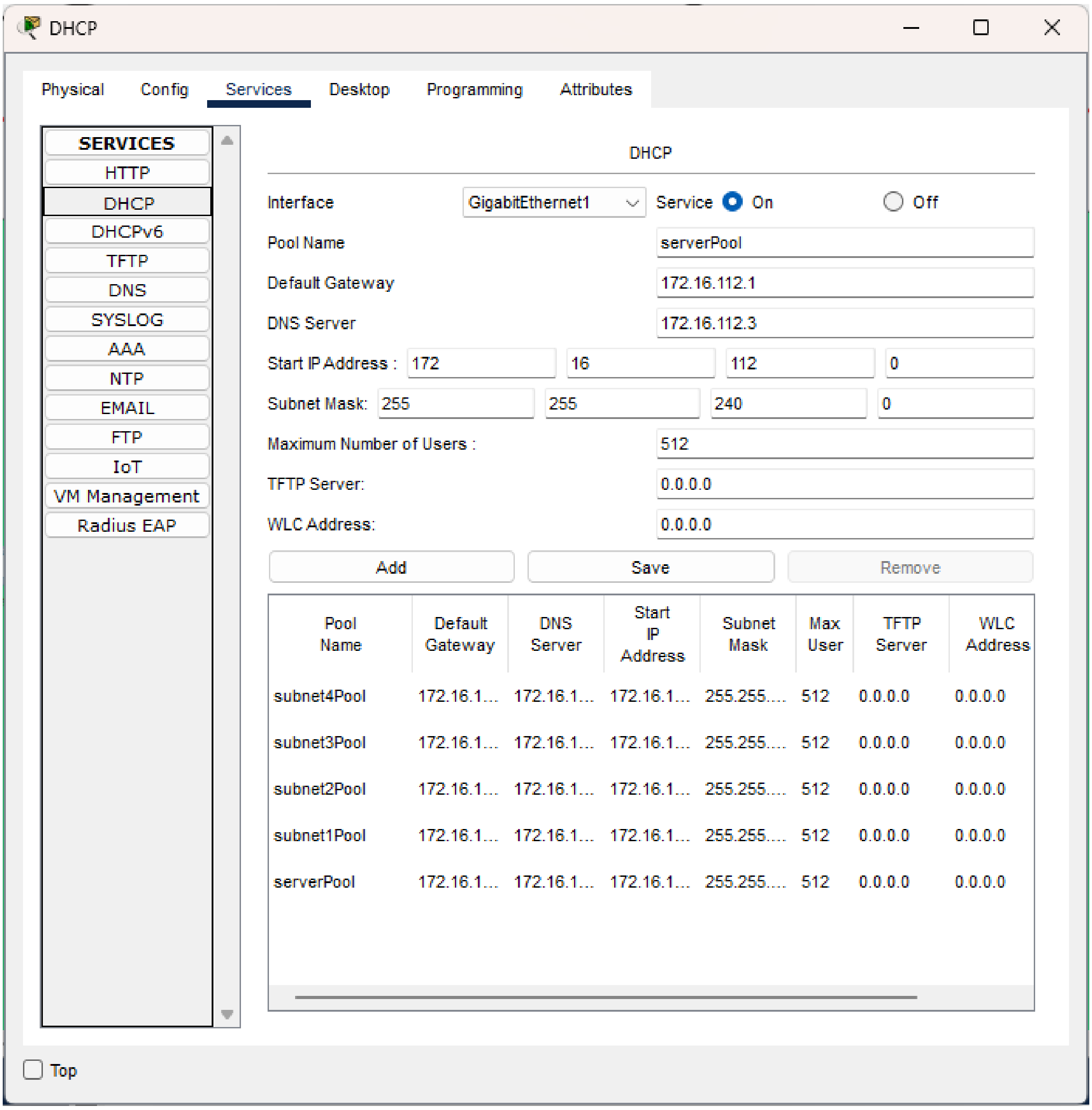


The above displays the commands to apply IP routing to router R144. A breakdown of the commands is:

- `enable`
  - grants root access to the router (not shown here).
- `configure terminal`
  - makes configuring the router possible.
- `interface [port name]`
  - makes configuring the port possible
- `ip helper-address [server address]`
  - `ip helper-address` is the command that enables DHCP relay, and points to the DHCP server.
  - `[server address]` is the address of the DHCP server.

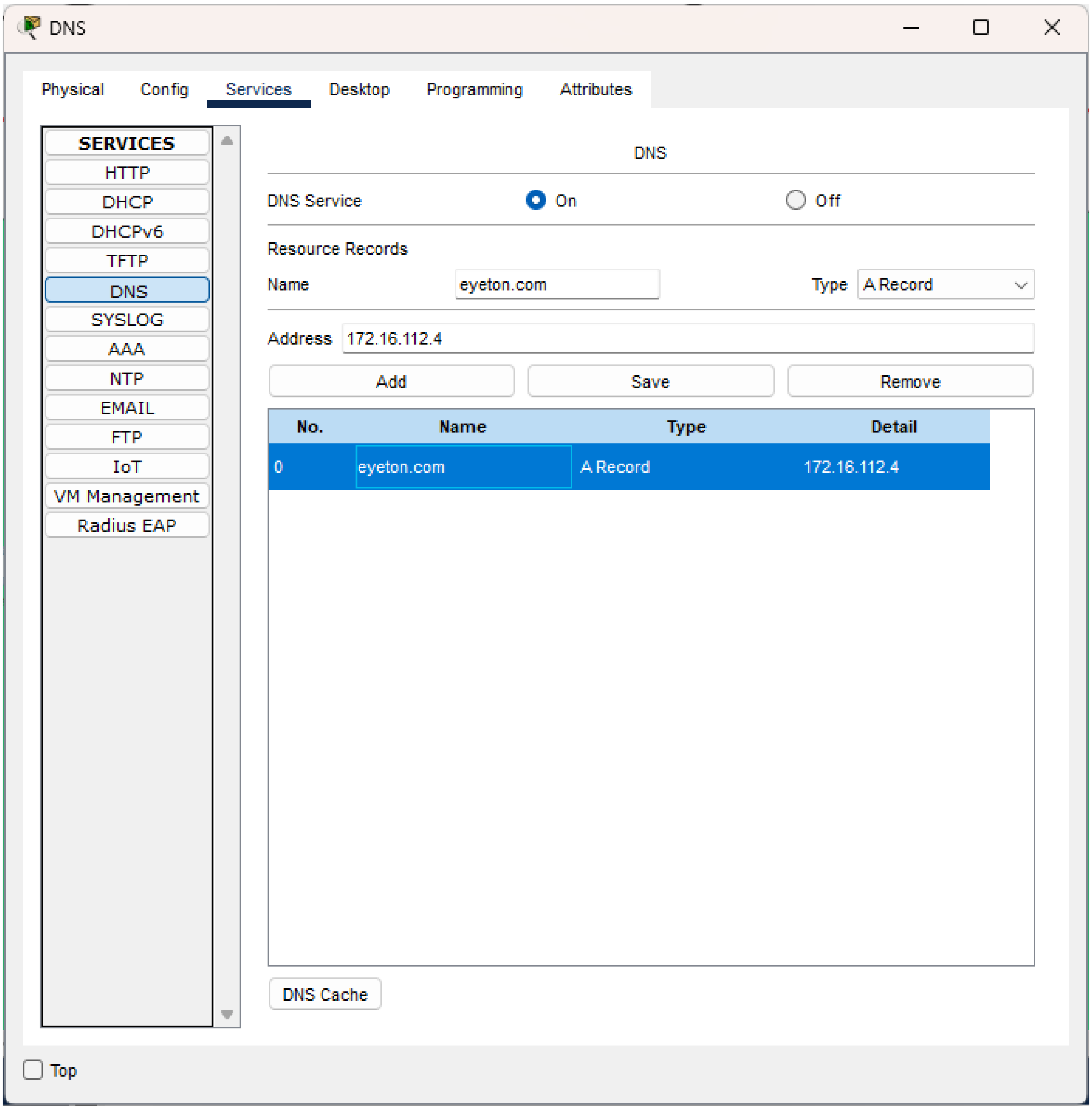
SERVER CONFIGURATION

DHCP SERVER



Above is the configuration screen for the DHCP server. As can be seen in the bottom pane, each subnetwork needs to have its dedicated address pool comprising of addresses that can be assigned to its connected devices.

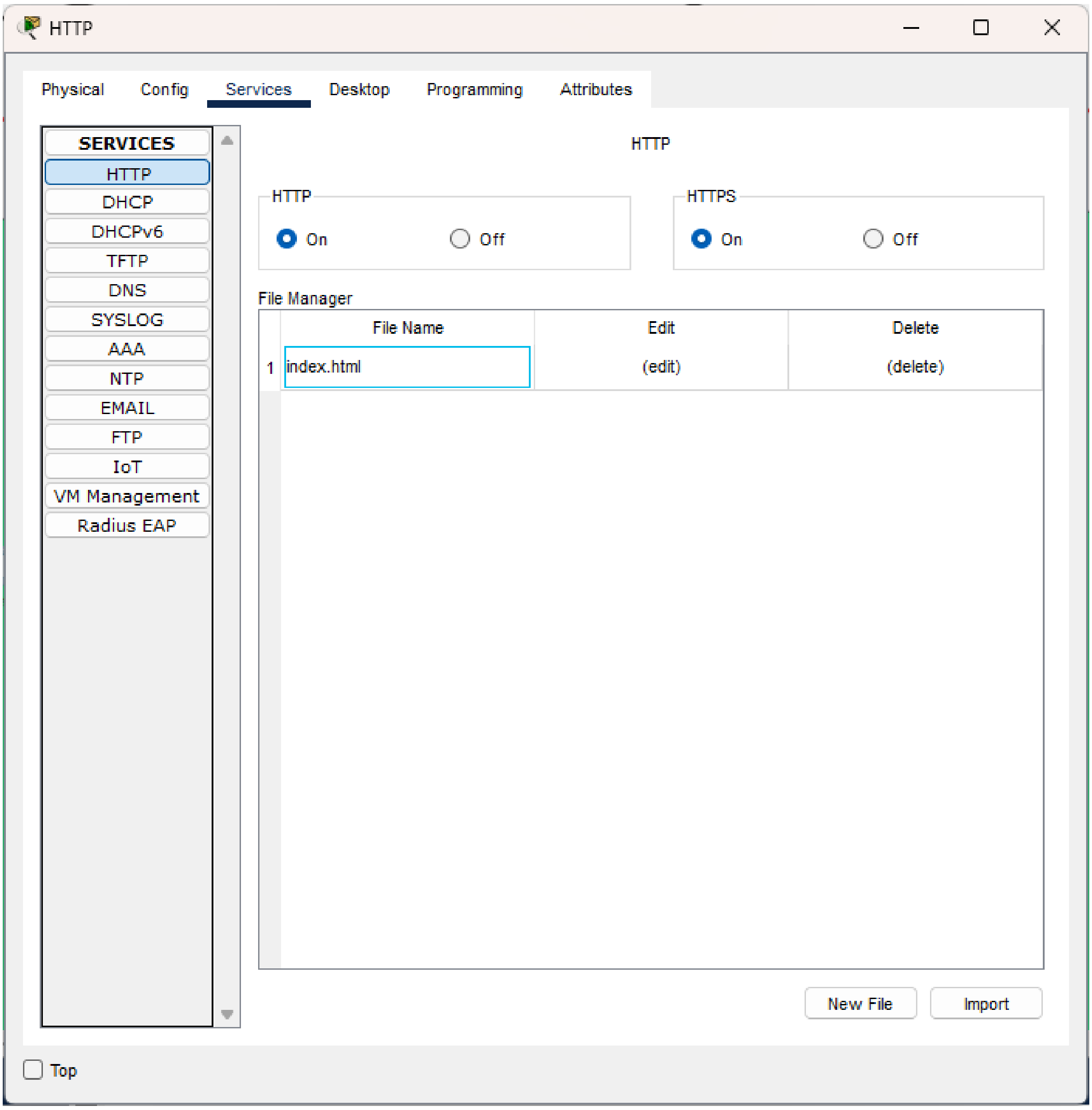
DOMAIN NAME SYSTEM SERVER



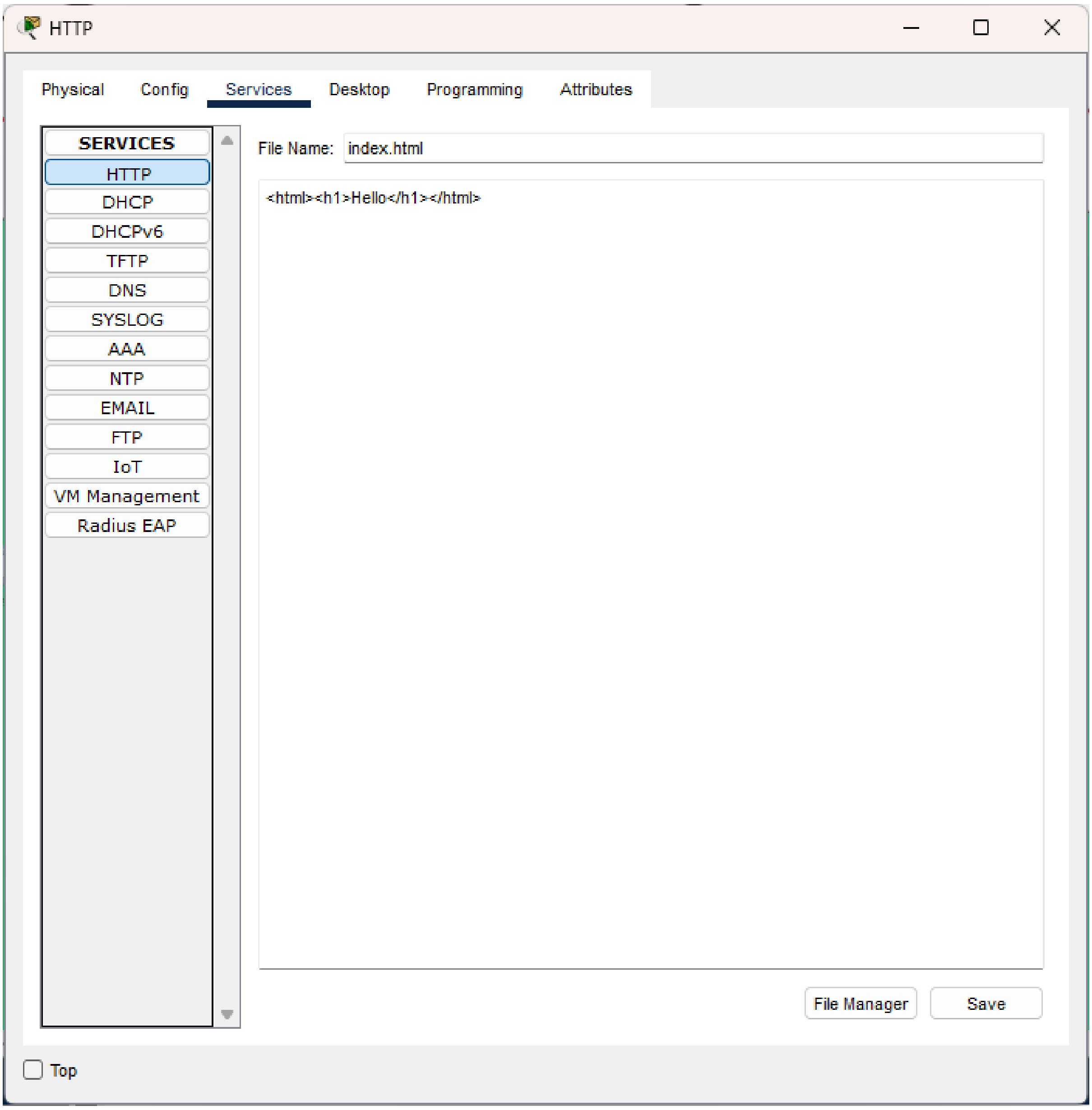
Above is the configuration screen for the DNS server.

- Name is for the web address of the site.
- Address is the IP address of the Web server hosting the website.

WEB SERVER



Above is the configuration screen of the Web server, and the files associated with the website.



Above is the configuration screen for the Web server; and displayed is the source code of the web page.

ALARM SIREN CONFIGURATION

AlarmSiren

Specifications | I/O Config | Physical | Config | Thing Editor | Programming | Attributes

Siren (Python) - main.py

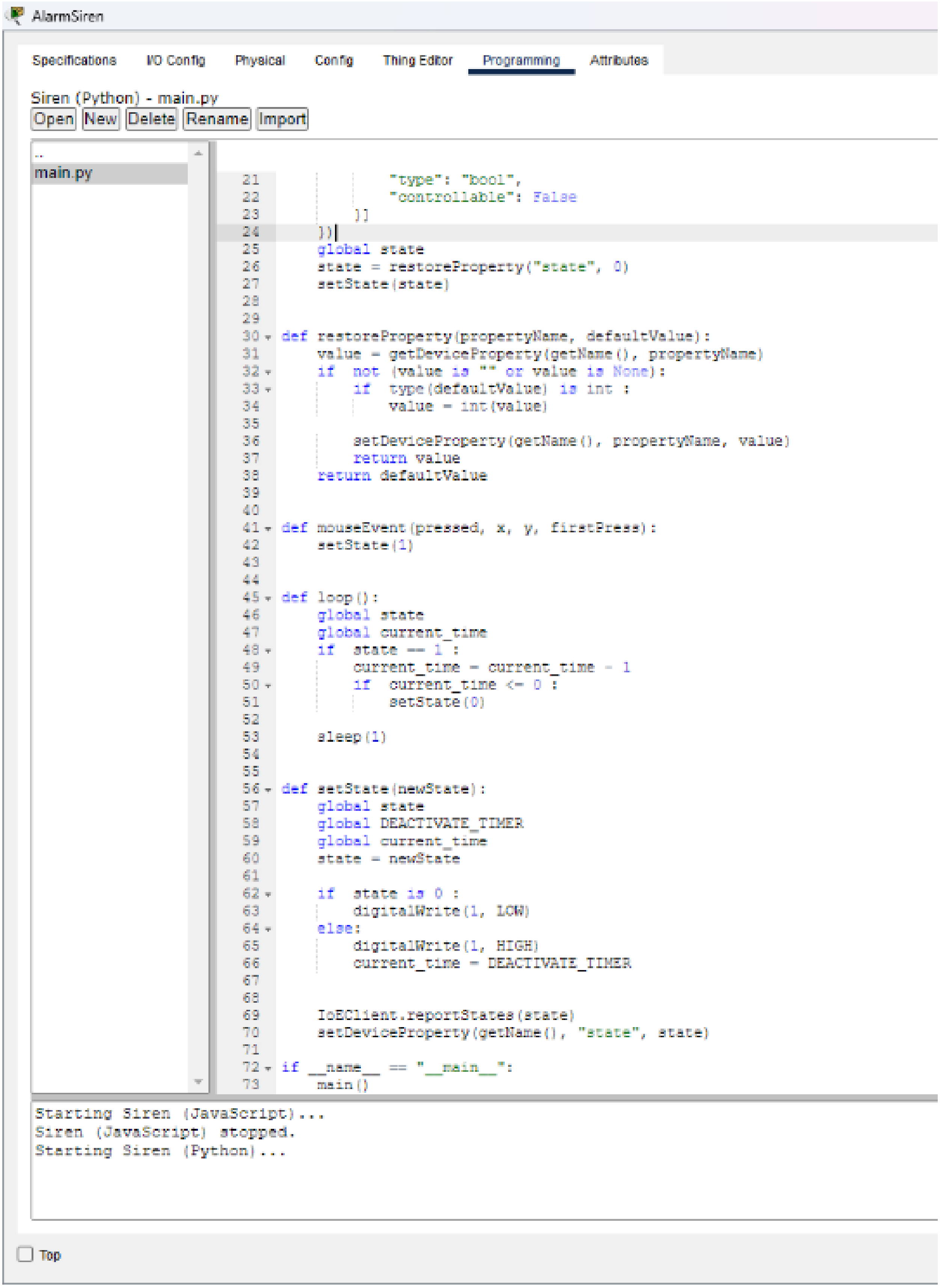
Open | New | Delete | Rename | Import

..
main.py

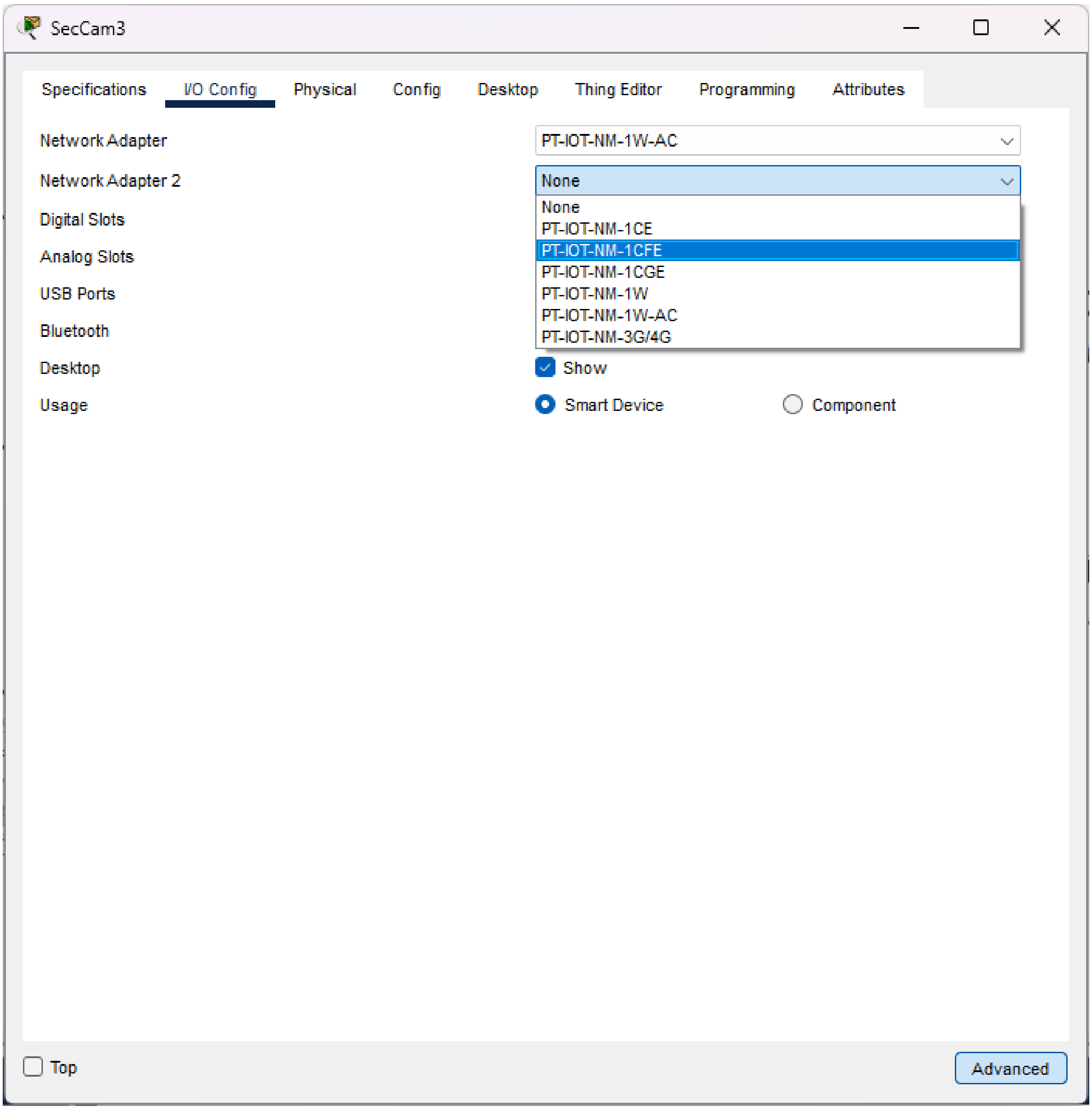
```python
from gpio import *
from time import *
from ioeclient import *
from physical import *
from bluetooth import *
import math

DEACTIVATE_TIMER = 5; # in seconds
state = 0
current_time = 0

def main():
    setup()
    while True:
        loop()
def setup():
    IoEClient.setup({
        "type": "AlarmSiren",
        "states": [{
            "name": "On",
            "type": "bool",
            "controllable": False
        }]
    })
    global state
    state = restoreProperty("state", 0)
    setState(state)


def restoreProperty(propertyName, defaultValue):
    value = getDeviceProperty(getName(), propertyName)
    if not (value is "" or value is None):
        if type(defaultValue) is int :
            value = int(value)

        setDeviceProperty(getName(), propertyName, value)
        return value
    return defaultValue


def mouseEvent(pressed, x, y, firstPress):
    setState(1)


def loop():
    global state
    global current_time
    if state == 1 :
        current_time = current_time - 1
        if current_time <= 0 :
            setState(0)

    sleep(1)
```

```
Starting Siren (JavaScript)...
Siren (JavaScript) stopped.
Starting Siren (Python)...
```

Top



Above is the configuration screen of the source code to the Alarm Siren. As can be seen, the alarm is to be triggered is motion is detected.

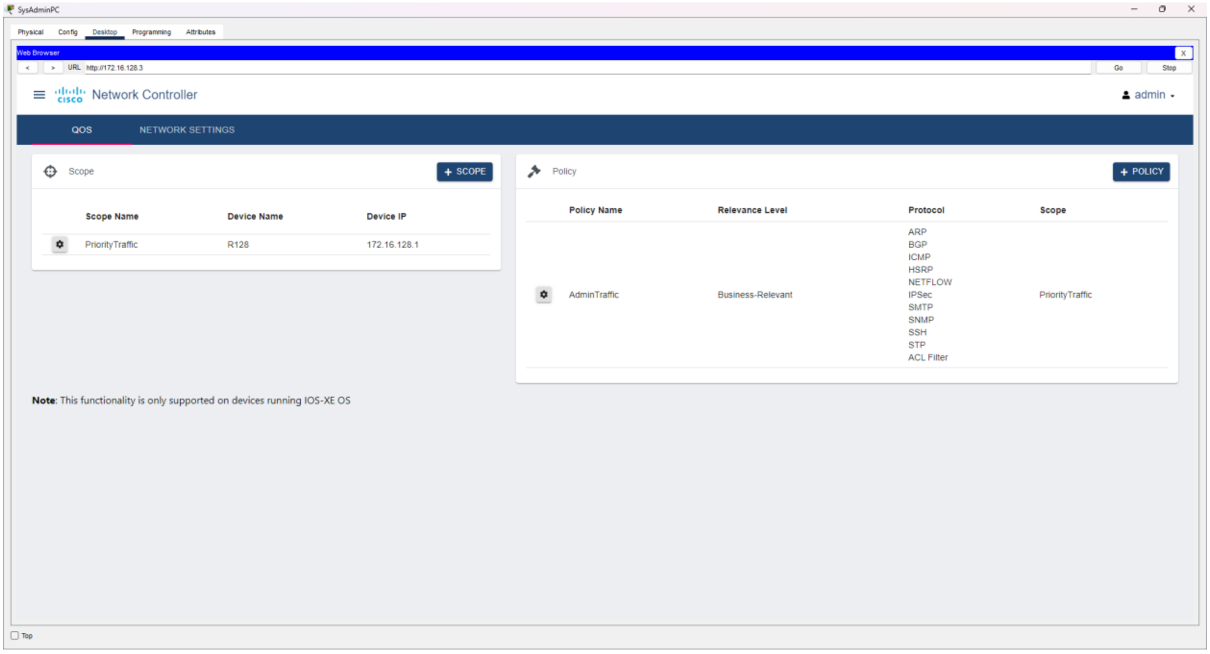


Above is the configuration screen for the Alarm Siren. A physical port has been added to connect it to the network through Ethernet cables. Process is repeated for the **Security Cameras.**

NETWORK CONTROLLER

The network controller provides a Graphical Interface to manage the network.

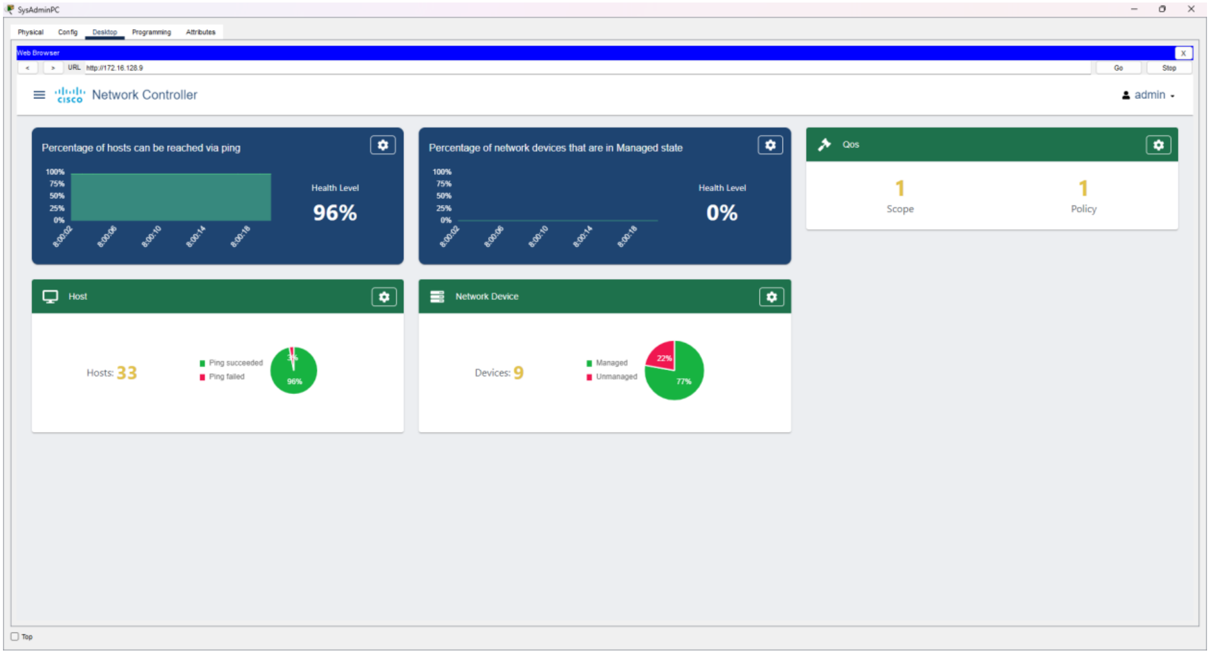
QUALITY OF SERVICE



Using the Network Controller, QoS has been implemented to give priority to the traffic coming from the System Administration, IT Support and Security Analyst departments.

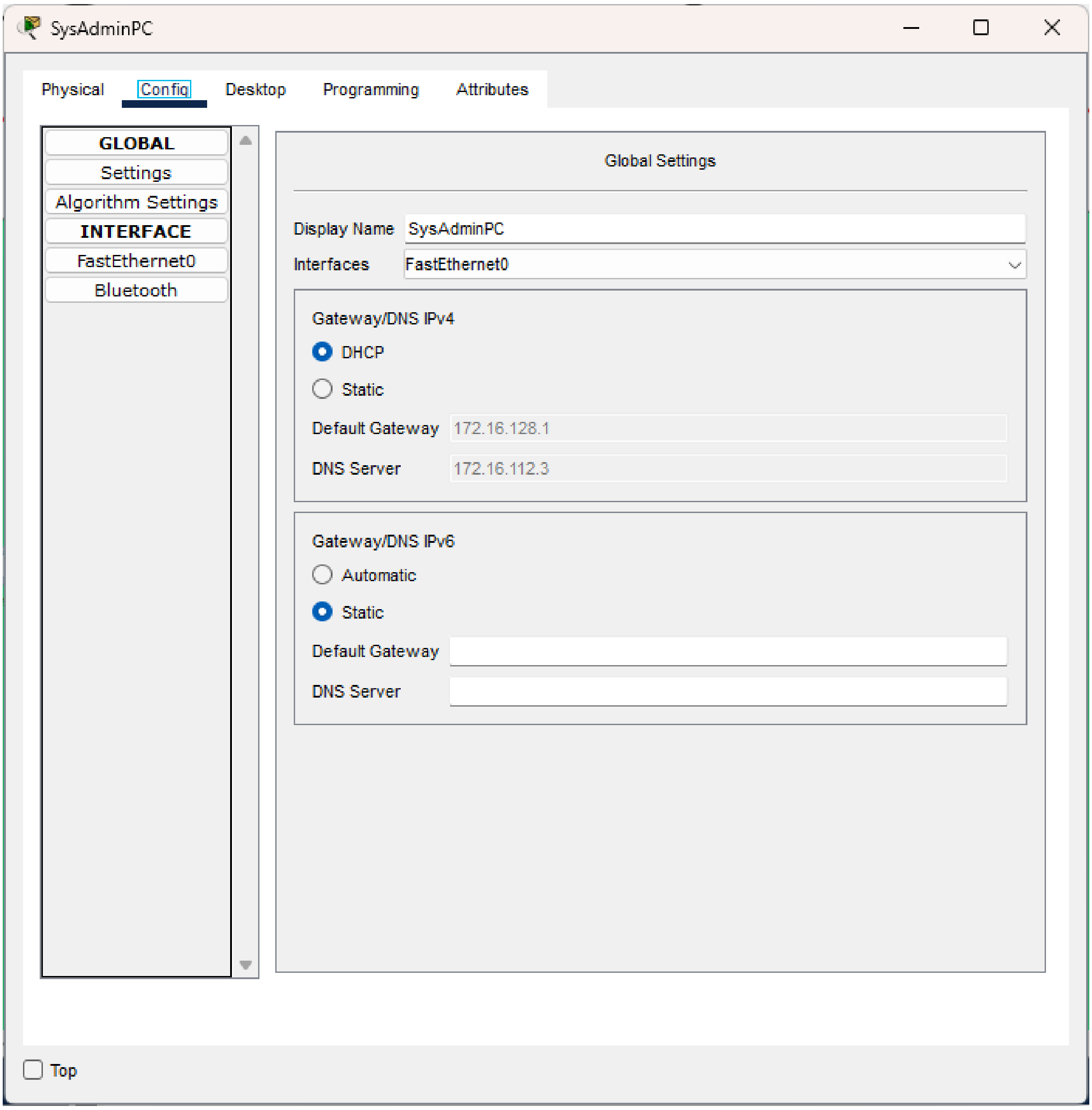
Quality of Service can also be implemented through the router itself, by entering the addresses of the devices that are concerned, which protocols are to be concerned, and the priority of traffic through the network. Alternatively, you can dynamically apply priority to all the traffic and protocols coming to and from an IP address, which automates process along the way.

NETWORK TESTING

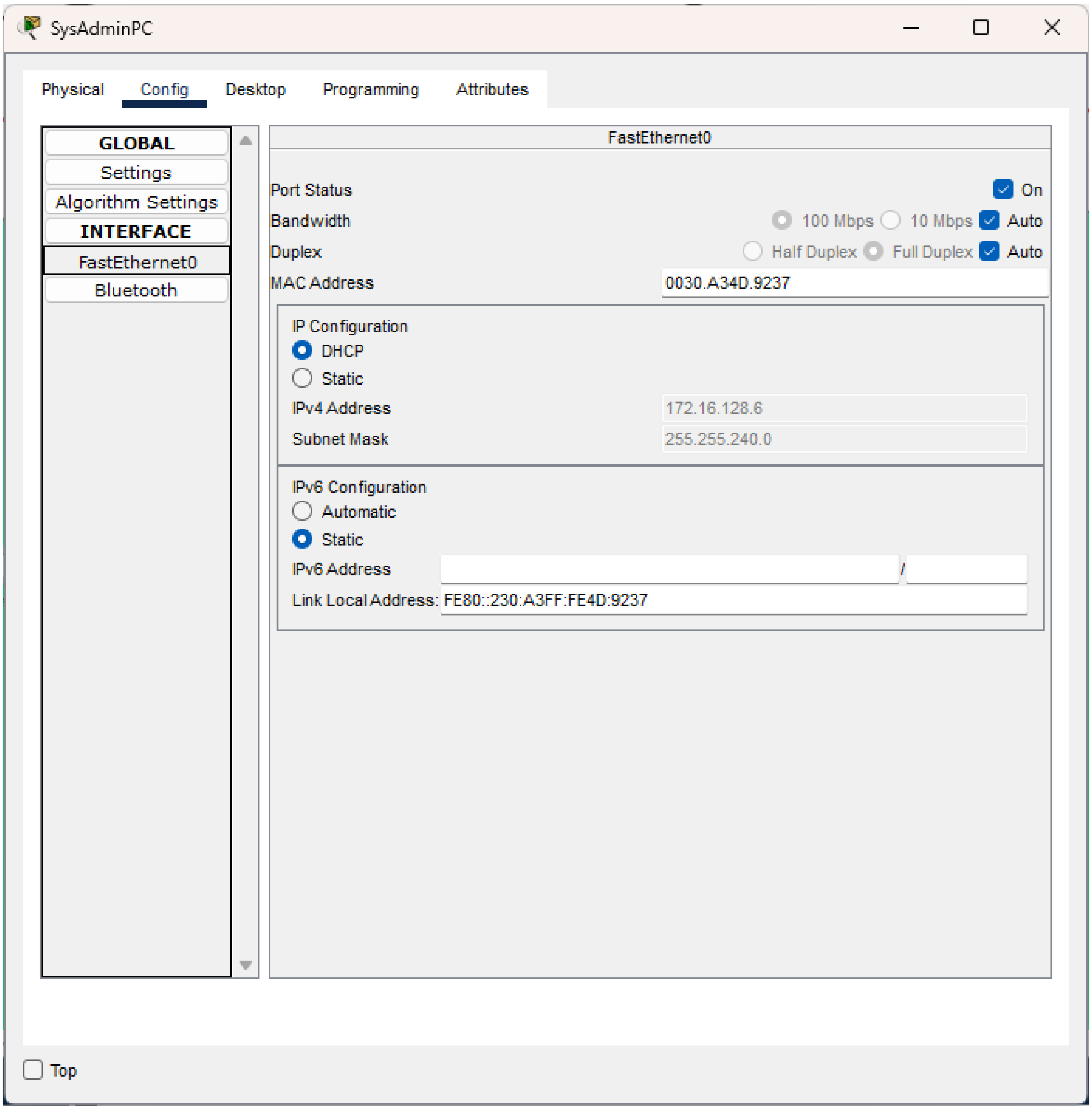


Above is the home screen of the Network Controller. It shows all the devices that are connected to the network, the QoS policies that has been applied, and the health of the network.

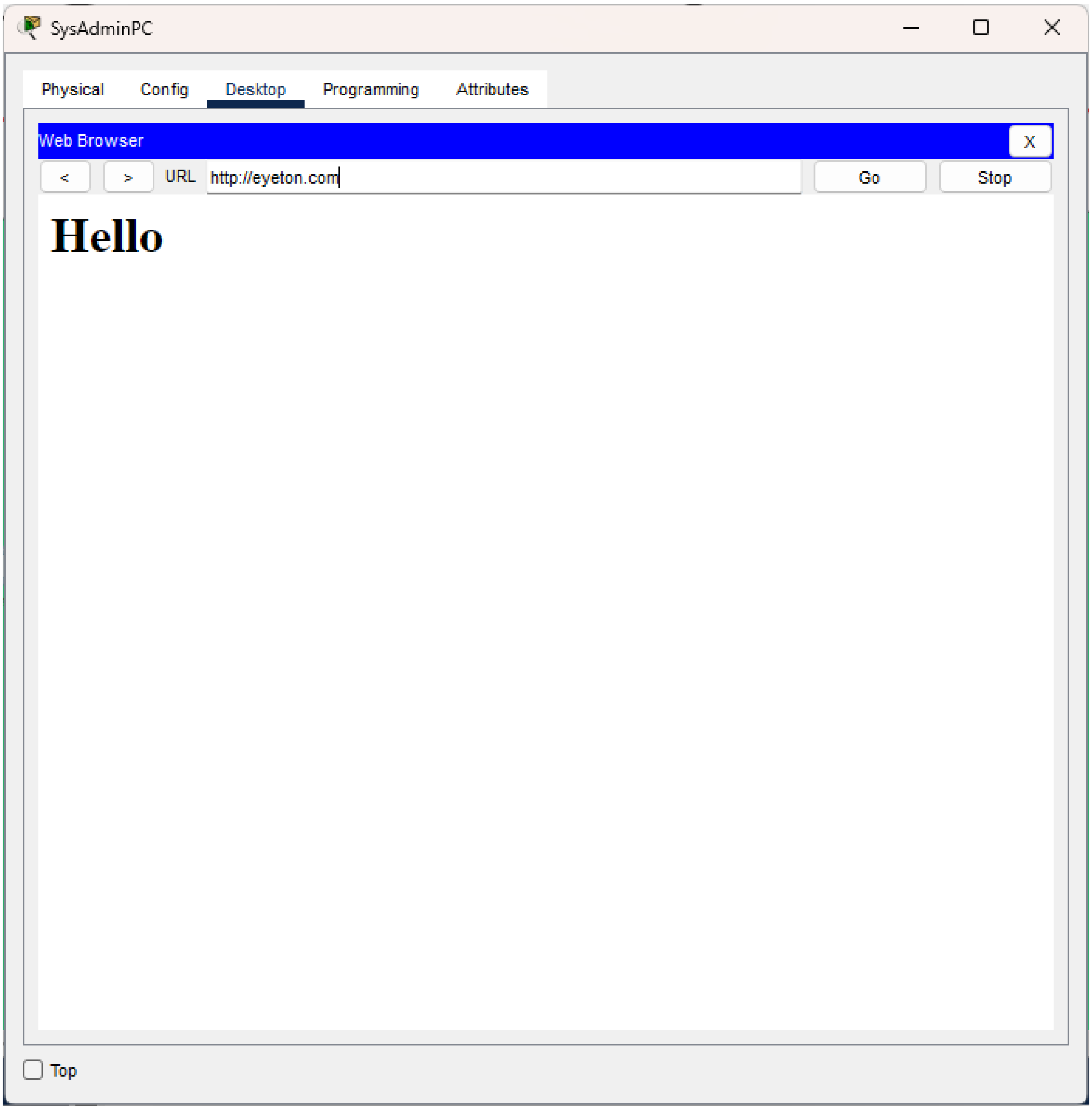
Note that the percentage of ping-able hosts should ideally be **100%** (which was achieved at a prior date, but unfortunately, documentation was not taken). This might be a bug in the software, because all the devices can be pinged through the CLI. This will need to be investigated further.



The above shows the DHCP resolution of the Default Gateway and DNS server.



The above shows the DHCP resolution of the IP address and subnet mask of a host.



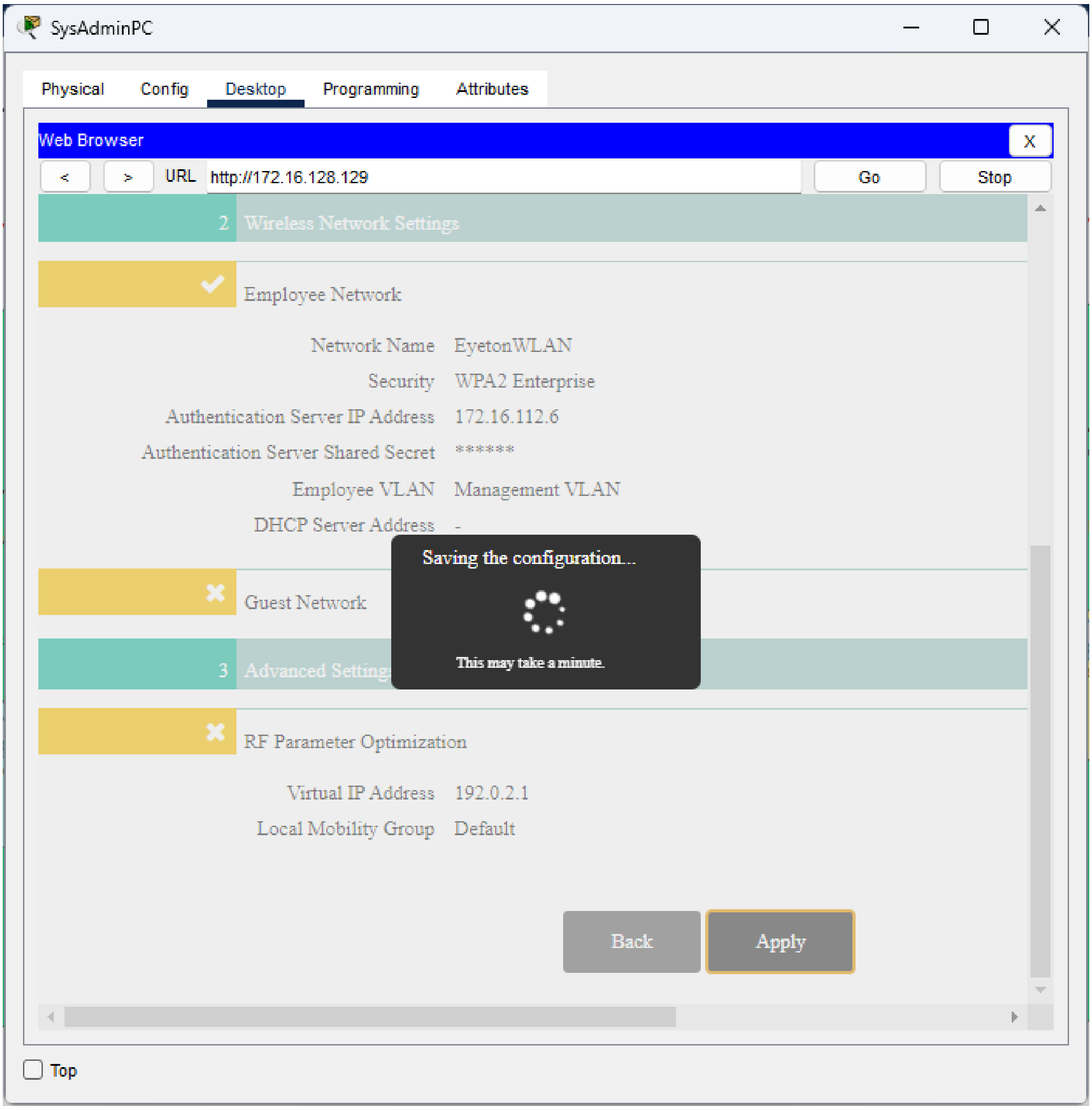
The above shows successfully accessing the home page of eyeton.com

NETWORK EVALUATION

The end goal of this assignment was to design, implement and test the network of an intelligence-based enterprise. The enterprise was divided into subnetworks for each department, each having their own respective end devices and the networking devices provided by Cisco Packet Tracer allowed us to successfully connect them together. Throughout the testing phase, we deduced that most devices were able to connect with each other, including devices from different subnets; however, the percentage of reachable hosts was still not 100%. Overall, the main objective was achieved even if the network was not fully perfect and we aim to provide further improvements to ensure that the network reaches its highest potential.

NOTES AND TROUBLESHOOTING

1) Initially, a network controller for the WLAN was to be implemented, however, through circumstances out of our hands, we were unable to do so – we were not able to save the configuration to the WLC, being stuck on this screen for what seemed like an eternity, multiple times.



2) Packet Capture Simulations had to always be manually ended as after the initial testing, the Network Controller would take over and send its share of STP packets to create active paths between each and every device, resulting in the simulation taking minutes to resolve.

3) Report focuses primarily on the Data Link and Network layers due to lack of time and limited testing capabilities of the Cisco Packet Tracer.
4) We did not go into too much depth in the implementation of the firewall, opting for the off-the-shelf configuration of the ASA6 5506-X as the security testing of the Cisco Packet Tracer is far too limited to truly stretch the firewall towards its limits.

RECOMMEDATIONS AND IMPROVEMENTS

1) Network Security Devices like NIDSs' and NIPS's will need to be implemented at a later time for Intrusion Detection and Intrusion Prevention purposes.
2) A SIEM system (Security Information and Event Management) will need to be implemented at a later time to log and analyse the activities within the network.
3) Vulnerability scanners will need to be implemented at a later time to scan our network for possible vulnerabilities and any other security weaknesses which might have slipped pass us.
4) A Network Access Control system will have to be implemented, so that devices are scanned for possible security vulnerabilities before they are allowed to connect to the network – especially in the case of wireless devices not handed out by the organisation.
5) For better DiD, VLAN Micro-Segmentation will need to be put in place to further partition the sub-networks for a better security posture.
6) RFID badge readers will need to be installed at each entry-way to act as another physical control.
7) A VPN server will need to be implemented, if ever the organisation is going to allow employees to work from home.
8) Redundancy will need to be implemented to prevent single-point of failures in the form of:
    a. Redundant switches and routers
    b. A backup server
    c. A hot site backup to ensure Business Continuity
    d. Redundancy in the data storage itself with the implementation of RAID (Redundant Array of Independent Disks)

CONCLUSION

The objective of this assignment was to set up an enterprise network and demonstrate its features using a network simulator software, which in this case was Cisco Packet Tracer. The availability of various networking devices that this software provides has allowed us to accurately design and implement all the departments that make up the entire network. These devices are now interconnected together to provide secure communication throughout the entire network. Additionally, the simulation mode of Cisco Packet Tracer has enabled us to perform the testing stage of this project, to prove that the network is indeed functional and further improvements of the network have been recommended in this report. Overall, this was a great opportunity to learn more about networks in general and how major companies can manage and set up their network. This is an experience that is to be implemented future projects

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